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Global prevalence of nontuberculous mycobacteria in adults with non-cystic fibrosis bronchiectasis 2006-2020:a systematic review and meta-analysis

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Global prevalence of nontuberculous mycobacteria in adults with non-cystic fibrosis bronchiectasis 2006-2020：a systematic review and meta-analysis

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ABSTRACT

Objectives There is an increasing interest in the prevalence of nontuberculous mycobacteria (NTM) in patients with non–cystic fibrosis (CF) bronchiectasis, which causes significant morbidity in adults. However, previously reported NTM prevalence in adults with non-CF bronchiectasis prior to 2006 was outdated and an update may guide clinical therapy by early eradication of the microbe in patients. We sought to accurately estimate the global prevalence of NTM in adults with non-CF bronchiectasis from 2006 to 2020 and determine the proportion of NTM species and subspecies in clinical patients.

Methods We systematically reviewed the existing reports from 2006 to 2020 according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. Meta-analysis with software R Statistics 3.6.3 using random effect model with a 95% confidence interval was performed to estimate the prevalence of NTM in patients with non–CF bronchiectasis. I^2 index and Q statistics were calculated to assess the heterogeneity, which was further studied by subgroup analyses. The proportions of patients tested positive for NTM were tested for normality using Shapiro-Wilk normality test in R.

Results 11 studies were included for the estimation. The global prevalence of NTM in adults with non-CF bronchiectasis from 2006 to 2020 was 10% with great variations primarily due to geographical location. Mycobacterium avium complex was the most common subspecies, followed by M. simiae and M. fortuitum.

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Conclusions The global prevalence of NTM in adults with non-CF bronchiectasis is 10%, which has been on the rise in recent years.

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Strengths and limitations of this study

- The patient data in this study was from 2006 to 2020 and therefore, this study significantly updated the estimate of the NTM prevalence in adults with non-cystic fibrosis bronchiectasis provided by a previous study with patient data from 1990 to 2006.
- This study provided a more accurate estimate of the NTM prevalence in adults with non-cystic fibrosis bronchiectasis with doubled number of included studies and ~10 times more patients for meta-analysis.
- Due to selection criteria, much data around the world must be excluded, and only one study from some continents such as Africa and America could be included for the estimation.

Key words: non-cystic fibrosis bronchiectasis, nontuberculous mycobacteria, prevalence, meta-analysis, subspecies.

INTRODUCTION

The incidence of non-cystic fibrosis (CF) bronchiectasis increased dramatically in

recent years, and it is the third most common respiratory disease after asthma and chronic obstructive pulmonary disease (COPD) [1–3]. The airways of patients with non-CF bronchiectasis are chronically colonized by a variety of pathologic microorganisms, such as yeasts, filamentous fungi, and nontuberculous mycobacteria (NTM) [4]. Currently It is widely believed that non-CF bronchiectasis and NTM are interrelated [5], and NTM causes human disease as opportunistic pathogens within a complex clinical context [6,7], such as COPD or CF [8]. Of these, the concomitant bronchiectasis is associated with a much greater relative risk of NTM infection than any other condition [9]. As a result, although the mechanism of NTM infection, as well as its impact on clinical outcomes, is not well understood [10], it is hypothesized that NTM infection may result in bronchiectasis [5].

The reported prevalence of NTM in the population with non-CF bronchiectasis varied widely among studies [11–14], and the NTM prevalence in adults with non-CF bronchiectasis from many studies might be overestimated due to the data originated from bronchiectasis medical/referral/registry centers, where NTM in respiratory secretions were routinely screened [14]. Meanwhile, global studies have discovered great geographic variation among species of NTM [15,16], which may account for the variability in the incidence of NTM pulmonary disease. These discoveries indicate that an accurate estimate of the global prevalence of NTM with subspecies in non-CF bronchiectasis is important to reveal the role of NTM in the disease and guide clinical treatments by dealing with the microbes in an earlier stage.

In a systematic review, Chu et al. reported that the prevalence of NTM infection

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was 9.3% [17]. Although this paper was published in 2014, all the clinical data were collected prior to 2006. The annually increasing prevalence of NTM [18–21] indicated that this data did not reflect the latest trend and should be updated. After in-depth analysis of the report by Chu et al, we found that the prevalence of NTM may not be accurate. First, bronchiectasis is more frequent in the people aged 40 years or older, but Chu's study included adolescence and children [17], causing an unknown percentage of NTM infections in the high-risk age groups [22,23]. Second, of all 8 studies included in Chu's study, two were from bronchiectasis medical/referral centers (Koh et al., 2005; Tabarsi et al., 2009), which may have overestimated the prevalence of NTM [24,25]. Koh et al. reported a prevalence of NTM as high as 30.0% [26]. However, many patients in the study by Koh et al were not bronchiectasis but bronchiolitis. Tabarsi, et al. reported a prevalence of NTM as 15.2%, but the cases might have included multidrug-resistant tuberculosis [27]. Thus, we believe that in this previous systematic review, only 6 studies should be included in the estimation of summarized prevalence of NTM in patients with bronchiectasis. We re-calculated the prevalence of NTM infection in patients with bronchiectasis of the 6 eligible studies and found that the accurate prevalence of NTM should be 5% (**Figure S1)**.

An updated and accurate estimate of the global prevalence of NTM infection is essential for a full understanding of the pathogenesis of non-CF bronchiectasis, enabling clinical doctors to find a complete cure of the disease. Herein we aim to determine the global prevalence of NTM in adults with non-CF bronchiectasis from 2006 to 2020 and identify the proportional relationships of NTM subspecies in clinical

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patients.

METHODS

We conducted a systematic review on NTM airway colonization in adults with non-CF bronchiectasis from 2006 to 2020 according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (PRISMA-P checklist; **Online Supplementary Appendix 1**) [28], and performed a meta-analysis on the global prevalence of NTM, as well as the proportion of their subspecies.

Search strategy and selection criteria

We searched the databases on Medline, Embase, Cochrane Library, Web of Science, and Google Scholar for articles published from 2006 to December 2020, using the keywords “nontuberculous mycobacter*”, “NTM”, “non-tuberculous mycobacter*”, “nontuberculous mycobacter*”, “atypical Mycobacter*”, “bronchiectasis”, “sputum”, “microbiome”, “bronchiectas*”, and “Kartagener*”. Databases were searched without language restrictions. We also reviewed the references cited by the selected articles to identify additional studies meeting our inclusion criteria. Two researchers (YCZ and WM) searched titles and abstracts and selected articles based on the inclusion criteria; any discrepancies were solved by consensus and with the help of a third reviewer (SWW).

We included all observational studies fulfilling the following criteria: 1) prospective or retrospective studies; 2) studies in adults (≥18 years of age) with bronchiectasis as defined by the authors; 3) studies estimating the effect size of the prevalence of NTM;

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4) sample size ≥40; 5) clinical patients or the vast majority of clinical patients in and after 2006; and 6) publications before 2021. We excluded studies if: 1) it only described NTM data unassociated with bronchiectasis; 2) it duplicated studies or records or did not calculate the prevalence of NTM in the patients of bronchiectasis; 3) participants included cystic fibrosis; 4) cultures included both NTM and tuberculous mycobacteria simultaneously; or, 5) case reports, commentary, or review articles.

Data extraction

Data was extracted by two reviewers (YCZ and WM) independently using the standard protocol and discrepancies were resolved through consultation with the third researcher (SWW). The following information of included studies were extracted: 1) last name of the first author, year of publication, country of the population studied; 2) time of study; 3) sample size; 4) mean age of patients; 5) data source; 6) method of cohort study, whether it is retrospective or prospective; 7) specimen source such as sputum or bronchoscopy specimen; 8) laboratory method of NTM tested; 9) definition of NTM positive; and, 10) infection rate of NTM in adults with non-CF bronchiectasis with the classification of NTM. For missing information, we communicated with the authors (**Supplementary Appendix Email S1**); if the information was unavailable after communications, this study was excluded. Two reviewers (YCZ and WM) evaluated the risk of bias in individual studies following the Cochrane Collaboration's tool (**Supplementary Appendix Table S1**) [29].

Quantitative analysis (meta-analysis)

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Software R Statistics 3.6.3 was employed for statistical analysis. The random effect model with a 95% confidence interval (CI) was adopted due to high heterogeneity in included studies. The I^2 index and Q statistic were calculated to assess study heterogeneity. The original prevalence of patients positive for NTM was tested for normality using the Shapiro-Wilk normality test in R. The results indicate that the prevalence requires transformation before pooling because they do not follow a normal distribution (W = 0.86764 p-value = 0.03116). The logit transformation of the original prevalence was then made and pooled [30]. The results of the fixed effect model were kept as a sensitivity analysis. Subgroup analyses were conducted by stratifying studies by features such as sample size, method of NTM examination, study type, and geographic location of the participants.

Patient and public involvement

There was no direct patient or public involvement in this review.

RESULTS

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1. Characteristics of eligible studies

The systematic review yielded 1898 articles. After eliminating duplicates and articles that did not match the inclusion criteria, 15 articles were considered eligible to be included in this systematic review (**Figure 1)**. The characteristics of the 15 studies were summarized in **Table 1**.

Figure 1 here.

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Table 1: Characteristics of studies included in the systematic review and meta-analysis

Author; Country; Year of publication	Time of sampling	Sample size	Patients with NTM	Mean age	Data Source	Study Design	Specimen source	Method of detection	Diagnosis of NTM positive	NTM [%]
1. McShane, et al.[18]; US; 2012	2009-2011	106	9	≥18 (Appendix V)	A US university Referral Center	RS	Sputum	Bacterial culture	American Thoracic Society criteria	8.5
2. Xu, et al.[19]; Shanghai, China; 2014	2009-2012	3857	431	47.4	Shanghai Pulmonary Hospital	RS	Sputum	MBC	At least one positive culture	11.2
3. Guan, et al.[32]; Guangzhou, China; 2015	2012-2013	144	5	44.6	Outpatient Respiratory Clinics	PS	Sputum	MBC	NTM culture positive	3.5
4. Kadowaki, et al.[33]; Japan; 2015	2008-2012	147	26	73	National Hospital Organization Matsue Medical Center	RS	sputum	MBC	Two samples were colonized with a species in a 1-year period	17.7
5. Máiz, et al. [12]; Spain; 2016	2012-2015	218	18	55.7	Teaching hospitals	PS	sputum	MBC	≥2 sputum samples culture positive	8.3
6. Izhakian，et al.[34]; Israel; 2016	2006-2014	339	29	64	Rabin Medical Center	RS	Bronchoalveolar lavage cultures	MBC	NTM culture positive	8.6
7. Faverio, et al. [20]; Italy; 2016	2006-2014	162	32	65	San Gerardo Hospital	PS	Bronchoscopic specimens and sputum	MBC	At least one isolated pathogen	19.8

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8.* Aksamit, *et al.*[11]; US; 2017	2008-2014	1,314	653	66	**Bronchiectasis Research Registry**	RS	Respiratory specimen	MBC	One or more NTM isolates in cultures	49.7
9.* Hsieh, et al.[35]; Taiwan, China; 2018	2005-2014	96	35	65.3	**Bronchiectasis Medical Center**	RS	sputum	MBC	At least two sputum samples positive	36.5
10. Sin, et al.[13]; South Korea; 2019	2005-2016	6957	1740	59.6	National University Hospital	RS	Respiratory specimen	MBC	Isolation of NTM at least once	25
11.* Amorim, et al.[36]; Portugal; 2019	2011-2017	186	27	54.7	**Bronchiectasis Referral Center**	PS	Sputum	MBC	NTM culture positive	14.5
12.* Visser, et al.[21] Australia; 2019	2016-2018	169	40	71	**Australian Bronchiectasis Registry**	PS	Respiratory specimen	MBC	One or more culture positive	23.6
13. Huang, et al.[37]; Taiwan, China; 2020	2002-2016	8385	304	65.5	Medical Record Database	RS	Sputum	MBC	NTM positive in two or more cultures	3.6
14. Darwish, et al.[38]; Egypt; 2020	2017-2018	40	3	55.2	University Hospital	PS	Sputum	PCR positive twice	NTM positive	7.5
15. Kwak, et al.[39]; South Korea; 2020	2011-2019	221	35	62	Hospital	PS	Sputum	MBC	NTM was isolated from the sputum for at least twice	15.8

Abbreviations: NA: Not Available; NTM: Nontuberculous mycobacteria; MBC: mycobacteria culture; AFB: acid fast bacillus smear; BAL: bronchoalveolar lavage; RS: Retrospective Study; PS: prospective Study; UK: United Kingdom; US: United States. PCR: Polymerase Chain Reaction. *Studies with source bias from bronchiectasis medical centers were also selected in initial analysis for a comparative analysis.

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2. Overall effects

2.1 Initial analysis of selected studies

Our sensitivity analysis of all the 15 included studies showed that the prevalence of NTM in adults with non-CF bronchiectasis was 14% (95% CI, 9%-20%) (**Figure 2**).

Figure 2 here.

Figure 2 indicated that three studies (Aksamit 2017, Hsieh 2018, and Visser 2019) were visually identified as outliers from the forest plot, whose prevalence (50%, 36% and 24%) were substantially higher and the confidence intervals had no or little overlaps with the CIs of other studies or the total effect size, indicating that these studies from bronchiectasis medical/registry/referral centers tremendously increased the estimated prevalence of NTM. Shteinberg and Aksamit believed that the estimates of NTM prevalence from bronchiectasis referral centers were exaggerated [40,41]. **Figure 2** indicates that including these 3 studies inevitably overestimated the percentage of NTM infection in adults with non-CF bronchiectasis and therefore, should be excluded. In addition, another study (Amorim 2019) was also from a bronchiectasis referral center. Thus, of all the 15 studies included, 4 from bronchiectasis referral centers addressed herein should be excluded from the data analysis to avoid overestimation of the prevalence of NTM infection.

2.2 An accurate estimate of the prevalence of NTM infection from 2006 to 2020

Thus, 11 independent studies were included in the meta-analysis for an accurate

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estimate of the prevalence of NTM infection in adults with non-CF bronchiectasis. These studies encompassed 20,576 patients with non-CF bronchiectasis and all the patients were ≥18 years old. The prevalence of NTM in adults with non-CF bronchiectasis ranged from 3.5% to 25%. Our sensitivity analysis showed that the prevalence of NTM infection in adults with non-CF bronchiectasis reduced from 14% to 10% (95% CI, 7%-15%) (**Figure 3**) after the 4 studies from bronchiectasis referral centers were removed. Results of the Chi-squared test for heterogeneity were 99%, suggesting a great degree of variations among the 11 included studies.

Figure 3 here.

3. Subgroup analysis

To understand the mechanisms of variations, we sub-grouped the included 11 studies based on 4 characteristics: sample sizes, the geographical locations of patients, study types, methods of NTM specimen examination (**Table 2**), and performed sensitivity analysis accordingly.

Table 2: Subgroup classification for data analysis

Study	Time of clinical registry	Number of NTM positive	Number of patients	Sample size	Study location	Study type	Culture method
1. McShane, 2012	2009-2011	10	106	< 200	North America	RS	MBC
2. Xu, 2014	2009-2012	431	3857	≥200	Asia	RS	MBC
3. Guan, 2015	2012-2013	5	144	< 200	Asia	PS	MBC

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4. Kadowaki, 2015	2008-2012	26	147	< 200	Asia	RS	MBC
5. Máiz, 2016	2012-2015	18	218	≥200	Europe	PS	MBC
6. Izhakian, 2016	2006-2014	29	339	≥200	Asia	RS	MBC
7. Faverio, 2016	2006-2014	32	162	< 200	Europe	PS	MBC
8. Sin, 2019	2005-2016	1740	6957	≥200	Asia	RS	MBC
9. Huang, 2020	2002-2016	304	8385	≥200	Asia	RS	MBC
10. Darwish, 2020	2017-2018	3	40	< 200	Africa	PS	PCR
11. Kwak, 2020	2011-2019	35	221	≥200	Asia	PS	MBC
Sum		2,643	20,576				

Abbreviations: RS: Retrospective Study; PS: Prospective Study; MBC: Mycobacteria Culture; AFB: Acid-fast Bacillus smear; PCR: Polymerase Chain Reaction.

3.1 The impact of sample size on the variations

First, our subgroup analysis on sample sizes indicates that the prevalence of NTM in the five studies with small sample (<200) was equal to that in the six studies with large sample (≥200) (10%, 95% CI 6%-18% vs. 10%, 95% CI 7%-15%) (**Figure 4**), indicating that sample size had limited impact on the variation of the global prevalence of NTM infection.

Figure 4 here.

3.2 The impact of geographical locations on the variation

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Second, we found that only one study was from North America and Africa, respectively, making the data on geographic locations incomparable. We noticed that the prevalence of NTM from the two continent is close to each other (8% vs. 7%), and the sample size of both studies were <200. Thus, we combined the two studies for data analysis (**Figure 5**). Our results indicated that in adult Europeans with bronchiectasis (two studies, 13%, 95% CI 7%-23%) the NTM prevalence was the highest, followed by adult Asians with non-CF bronchiectasis (seven studies, 10%, 95% CI 6%-17%), and adult Americans and Africans with non-CF bronchiectasis (two studies, 8%, 95% CI 5%-14%). Moreover, the forest plot clearly indicated that the proportion within a continent of Asia or Europe varied greatly, and the 95% CI among countries rarely overlapped, indicating that geographic locations account for the variations of the prevalence of NTM.

Figure 5 here.

3.3 The impact of study types on the variation

Third, our subgroup analysis on study types (retrospective and prospective) found that the estimated NTM prevalence from the six prospective studies was 1% higher than that of the five retrospective studies (11%, 95% CI 6%-17% vs. 10%, 95% CI 6%-17%) (**Figure 6**). Although the Chi-squared test indicated better homogeneity among prospective studies, the distinction bore no significance in statistics, indicating study type was not the cause of the variations of NTM prevalence.

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Figure 6 here.

3.4 The impact of methods of specimen examination on the variation

Lastly, the subgroup analysis on the methods of specimen examination indicated that, although the NTM prevalence in the ten studies using culturing mycobacteria (MBC) was 4% higher than the study using Polymerase Chain Reaction (PCR) (11%, 95% CI 7%-16% vs. 7%, 95% CI 2%-21%) (**Figure 7**), the difference was not statistically significant, indicating methods of specimen examination has very little impact on variation of the NTM prevalence.

Figure 7 here.

3.5 Results of subgroup analyses

Global inspection of the results of subgroup analyses suggested that the variations among the 11 included studies was not primarily due to sample sizes, study types, and specimen examination methods, but geographic locations. Further studies on regional difference, such as climate, soil, water, culture, may help to understand the epidemiology of NTM infection in adults with non-CF bronchiectasis.

4. Subspecies analysis

Finally, we estimated the prevalence of NTM subspecies in non-CF adult bronchiectasis patients of the included 11 studies (**Table 3**). Four (Xu 2014, Guan 2015, Sin 2019, and Huang 2020) did not report the number of confirmed cases or

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specimens of NTM subspecies and therefore were not included in this analysis (**Table S2**). The results showed that the most widely detected NTM subspecies was MAC (68.38%, 95% CI 49.08%-87.68%), followed by *M. simiae* (12.4%, 95% CI 0%-25.17%) and *M. fortuitum* (7.95%, 95% CI 1.17-14.72).

Table 3: The proportion of subspecies in all NTM positive specimens

NTM subspecies	No. of studies	Proportion (95% CI) [%]	Effect model	Method of data transformation	Test of heterogeneity		
					Q	P-value	I^2 [%]
MAC	7	68.38 (49.08-87.68)	Random	PRAW	67.91	<0.01	91.2
M. abscessus	4	6.25 (1.93-10.58)	Fixed	PRAW	3.19	0.36	6.1
M. chelonae	4	6.29 (1.39-11.2)	Fixed	PRAW	0.10	0.99	0
M. gordonae	3	1.49 (3.05-17.93)	Fixed	PRAW	0.23	0.89	0
M. fortuitum	3	7.95 (1.17-14.72)	Fixed	PRAW	0.19	0.91	0
M. simiae	3	12.4 (0-25.17)	Random	PRAW	5.43	0.07	63.1
M. kansasii	2	3.68 (0-9.11)	Fixed	PRAW	0.34	0.56	0
Undetermined	2	11.9(1.82-21.99)	Fixed	PRAW	0.64	0.42	0
M. xenopi	1	7.69 (1/13)	-	-	-	-	-
M. lentiflavum	1	5.26 (1/19)	-	-	-	-	-

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M. shimoidei	1	3.03 (1/33)	-	-	-	-	-

Abbreviations: MAC: *Mycobacterium Avium Complex*; M. chelonae: Mycobacterium chelonae; M. kansasii: Mycobacterium kansasii; M. abscessus: Mycobacterium abscessus; M. fortuitum: Mycobacterium fortuitum; M. gordonae: Mycobacterium gordonae; M. shimoidei: Mycobacterium shimoidei; M. avium intracellulare: Mycobacterium avium intracellulare; M. lentiflavum: Mycobacterium lentiflavum; M. simiae: Mycobacterium simiae; M. xenopi: Mycobacterium xenopi; M. peregrinum: Mycobacterium peregrinum; PRAW=Raw data; PLOGIT=Logit transformation; PFT=Freeman-Tukey double arcsine transformation.
Notes: Proportion in each study = number of specimens tested positive for a specific subspecies/total NTM-positive specimen in the study.

DISCUSSION

Main Findings

An updated, accurate estimate of global prevalence of NTM infection is of clinical significance. To accurately estimate the most recent global prevalence of NTM in adults with non-CF bronchiectasis and to determine the proportion of NTM species and subspecies in these patients, we performed a systematic review and included 11 studies from multiple geographic locations in the meta-analysis. Our results showed that, the global prevalence of NTM infection in adults with non-CF bronchiectasis from 2006 to 2020 was 10% (95% CI, 7%-15%). There were major variations in the prevalence among studies [11–14], primarily driven by geographic differences. Moreover, we found that MAC was the most common subspecies (58%), followed by *M. simiae* (12.40%) and *M. fortuitum* (7.95%).

Our subgroup analyses indicate that, of all the 4 factors we investigated, the great variation of the prevalence of NTM among studies was primarily due to geographic difference, whereas sample size, study type, and detection method only slightly affect the estimates. The airway colonization of NTM in Europe is the highest

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(13%), followed by Asia (10%), North America (8%) and Africa (7%). Interestingly, the prevalence of NTM among countries or within the same continent, or even among regions within the same country also varied greatly (**Figure 5**). The mechanisms for this geographic variation remain elusive, and unknown factors may contribute to the high prevalence of NTM in some countries or regions. It was speculated that environment factor such as climate, soil, and water [6–8,43], and regional differences in techniques [44] may account for the geographic variations of NTM prevalence. Further studies on the geographic variations may contribute to the prevention of NTM in adults with non-CF bronchiectasis.

Comparison with literature

After excluding two invalid studies (Koh et al., 2005; Tabarsi et al., 2009) in a previous study by Chu et al. in 2014, the real global prevalence of NTM infections from 1990 to 2006 was 5% (**Figure S1**). Herein our systematic review and meta-analysis results indicated that the global prevalence of NTM infections for the period from 2006 to 2020 was 10%. The doubled percentage (10% vs. 5%) implicated the gradually increasing prevalence of NTM infections in adults with non-CF bronchiectasis over the past 15 years. Moreover, our initial analysis suggested that four studies from bronchiectasis referral center did overestimate the prevalence of NTM infection. Thus, they should not be included for the estimate of the global prevalence of NTM. Both Shteinberg and Aksamit also believed that the estimates of NTM prevalence from bronchiectasis referral centers were exaggerated [14,42]. Significantly, the data we presented herein encompassed a much larger number of

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non-CF bronchiectasis patients (20,576 vs. 1,492) from more geographic locations (11 vs. 6) than the study of Chu et al in 2014, making it more representative and more clinically relevant.

Implications for research and practice

The increasing interest on NTM infection in non-CF bronchiectasis worldwide in recent years indicates the microbes may play a role in the pathogenesis of the disease [20]. In general, of all patients with a positive culture, only 25~60% meet the criteria for NTM pulmonary disease [45]. Our data showed that up to 50%, or even up to 90% of the total NTM infections in bronchiectasis patients were subspecies *MAC*, similar to the data previously estimated [26,46,47]. Significantly, the isolation of *M. kansasii*, *M. szulgai*, and *M. malmoense* were recognized as causative agents in most patients, whereas *M. gordonae, M. terrae,* and *M. fortuitum* were less virulent and they were usually believed to be contaminants rather than causative agents [45,48]. Therefore, the isolation of a NTM subspecies has been demonstrated to be clinically important [48,49]. However, a diagnosis of infection in lung disease does not dictate the initiation of antibiotic therapy against NTM species. Instead, the healthcare team should consider the potential risks and benefits for individual patients of a prolonged course of treatment with multiple antibiotics [50].

Strengths and limitations

In the paper by Chu et al. in 2014, all the clinical data was collected prior to 2006. In contrast, the vast majority of the clinical data collected in this paper was between 2006 and 2020. There were no data and time overlaps between the two studies and thus, results from

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this article may represent the most recent trend of NTM prevalence in adults with non-CF bronchiectasis. Moreover, in contrast to the paper in 2014, this study almost doubled the number of valid studies (11 vs. 6), analyzed the data with much larger sample size (20,576 vs. 1,492) from more geographic locations (11 vs. 6) and we take source bias into our consideration. Hence, we were able to present an updated and more accurate estimate of the prevalence of NTM in adults with non-CF bronchiectasis, which may be significant to direct clinical treatments.

However, there were several limitations as well. First, only one study matched the inclusion criteria from Africa, North America, and Oceania, respectively, indicating that the data for these continents was less representative. Second, only one study, in which PCR was employed to detect NTM species, meets the selection criteria, making the detection methods incomparable. Third, 5 studies did not report the number of specimens of NTM subspecies and therefore were not included in the data analysis for subspecies. Fourth, the percentage presented herein (10%) is an overall estimate for the global prevalence of NTM in adults with non-CF bronchiectasis 2006-2020. As per the prevalence of NTM in a specific country or region, retrospective and prospective studies are still needed for a customized and accurate estimate. Based on the analysis and issues presented in this study, other researchers may replenish the data and provide a more accurate and up-to-date estimation in the future.

CONCLUSIONS

Our systematic review and meta-analysis suggest that the global prevalence of NTM

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in adults with non-CF bronchiectasis from 2006 to 2020 is about 10%, with major variations in the estimated prevalence primarily driven by geographical locations. *MAC* is the most common organism in non-CF bronchiectasis, followed by *M. simiae* and *M. fortuitum*.

Abbreviations: NTM，nontuberculous mycobacteria；CF，non-cystic fibrosis；COPD, chronic obstructive pulmonary disease；CI, confidence interval；NA: Not Available; MBC: mycobacteria culture; AFB: acid fast bacillus smear; BAL: bronchoalveolar lavage; RS: retrospective study; PS: prospective study; UK: United Kingdom; US: United States. PCR: polymerase chain reaction

Author contributions YC, SP, and SWW contributed to the conception and design of the study. YC, WM, SP, and SWW made substantial contributions to the acquisition of the data, data entry, or data analysis, and data interpretation. YC, SP, and SWW were all involved in the drafting of the manuscript, or the revisions made to the manuscript. All authors have given final approval of the version of the manuscript submitted for publication and agree to be accountable for all aspects of the work.

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Competing statement All authors have no conflict of interest to report.

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Patient consent Not required.

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Data sharing statement All papers included in this study are publicly published articles which are accessible online.

Online Supplementary Appendix 1. PRISMA-P checklist; **Appendix 2:** Supplementary tables, figure, and communication email.

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Table 1: Characteristics of studies included in the systematic review and meta-analysis

Author; Country; Year of publication	Time of sampling	Sample size	Patients with NTM	Mean age	Data Source	Study Design	Specimen source	Method of detection	Diagnosis of NTM positive	NTM [%]
1. McShane, et al. [18]; US; 2012	2009-2011	106	9	≥18 (Supporting Document 1)	A US university Referral Center	RS	Sputum	Bacterial culture	American Thoracic Society criteria	8.5
2. Xu, et al. [19]; Shanghai, China; 2014	2009-2012	3857	431	47.4	Shanghai Pulmonary Hospital	RS	Sputum	MBC	At least one positive culture	11.2
3. Guan, et al. [32]; Guangzhou,	2012-2013	144	5	44.6	Outpatient Respiratory	PS	Sputum	MBC	NTM culture positive	3.5

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China; 2015					Clinics					
4. Kadowaki, et al. [33]; Japan; 2015	2008-2012	147	26	73	National Hospital Organization Matsue Medical Center	RS	sputum	MBC	Two samples were colonized with a species in a 1-year period	17.7
5. Máiz, et al. [12]; Spain; 2016	2012-2015	218	18	55.7	Teaching hospitals	PS	sputum	MBC	≥2 sputum samples culture positive	8.3
6. Izhakian，et al. [34]; Israel; 2016	2006-2014	339	29	64	Rabin Medical	RS	Bronchoalveolar lavage	MBC	NTM culture positive	8.6

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					Center		cultures			
7. Faverio, et al. [20]; Italy; 2016	2006-2014	162	32	65	San Gerardo Hospital	PS	Bronchoscopic specimens and sputum	MBC	At least one isolated pathogen	19.8
8.* Aksamit, *et al.* [11]; US; 2017	2008-2014	1,314	653	66	**Bronchiectasis Research Registry**	RS	Respiratory specimen	MBC	One or more NTM isolates in cultures	49.7
9.* Hsieh, et al. [35]; Taiwan, China; 2018	2005-2014	96	35	65.3	**Bronchiectasis Medical**	RS	sputum	MBC	At least two sputum samples positive	36.5

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					Center				
10. Sin, et al. [13]; South Korea; 2019	2005-2016	6957	1740	59.6	National University Hospital	RS	Respiratory specimen	MBC	Isolation of NTM at least once	25
11.* Amorim, et al. [36]; Portugal; 2019	2011-2017	186	27	54.7	**Bronchiectasis Referral Center**	PS	Sputum	MBC	NTM culture positive	14.5
12.* Visser, et al. [21] Australia; 2019	2016-2018	169	40	71	**Australian Bronchiectasis Registry**	PS	Respiratory specimen	MBC	One or more culture positive	23.6

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13. Huang, et al. [37]; Taiwan, China; 2020	2002-2016	8385	304	65.5	Medical Record Database	RS	Sputum	MBC	NTM positive in two or more cultures	3.6
14. Darwish, et al. [38]; Egypt; 2020	2017-2018	40	3	55.2	University Hospital	PS	Sputum	PCR positive twice	NTM positive	7.5
15. Kwak, et al. [39]; South Korea; 2020	2011-2019	221	35	62	Hospital	PS	Sputum	MBC	NTM was isolated from the sputum for at least twice	15.8

Abbreviations: NA: Not Available; NTM: Nontuberculous mycobacteria; MBC: mycobacteria culture; AFB: acid fast bacillus smear; BAL: bronchoalveolar lavage; RS: Retrospective Study; PS: prospective Study; UK: United Kingdom; US: United States. PCR: Polymerase Chain Reaction. *Studies with source bias from bronchiectasis medical centers were also selected in initial analysis for a comparative analysis.

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Table 2: Subgroup classification for data analysis

Study	Time of clinical registry	Number of NTM positive	Number of patients	Sample size	Study location	Study type	Culture method
1. McShane, 2012	2009-2011	10	106	< 200	North America	RS	MBC
2. Xu, 2014	2009-2012	431	3857	≥200	Asia	RS	MBC
3. Guan, 2015	2012-2013	5	144	< 200	Asia	PS	MBC
4. Kadowaki, 2015	2008-2012	26	147	< 200	Asia	RS	MBC
5. Máiz, 2016	2012-2015	18	218	≥200	Europe	PS	MBC
6. Izhakian, 2016	2006-2014	29	339	≥200	Asia	RS	MBC
7. Faverio, 2016	2006-2014	32	162	< 200	Europe	PS	MBC

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8. Sin, 2019	2005-2016	1740	6957	≥200	Asia	RS	MBC
9. Huang, 2020	2002-2016	304	8385	≥200	Asia	RS	MBC
10. Darwish, 2020	2017-2018	3	40	< 200	Africa	PS	PCR
11. Kwak, 2020	2011-2019	35	221	≥200	Asia	PS	MBC
Sum		2,643	20,576				

Abbreviations: RS: Retrospective Study; PS: Prospective Study; MBC: Mycobacteria Culture; AFB: Acid-fast Bacillus smear; PCR: Polymerase Chain Reaction.

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Table 3: The proportion of subspecies in all NTM positive specimens

NTM subspecies	No. of studies	Proportion (95% CI) [%]	Effect model	Method of data transformation	Test of heterogeneity		
					Q	P-value	I^2 [%]
MAC	7	68.38 (49.08-87.68)	Random	PRAW	67.91	<0.01	91.2
M. abscessus	4	6.25 (1.93-10.58)	Fixed	PRAW	3.19	0.36	6.1
M. chelonae	4	6.29 (1.39-11.2)	Fixed	PRAW	0.10	0.99	0
M. gordonae	3	1.49 (3.05-17.93)	Fixed	PRAW	0.23	0.89	0
M. fortuitum	3	7.95 (1.17-14.72)	Fixed	PRAW	0.19	0.91	0
M. simiae	3	12.4 (0-25.17)	Random	PRAW	5.43	0.07	63.1
M. kansasii	2	3.68 (0-9.11)	Fixed	PRAW	0.34	0.56	0
Undetermined	2	11.9(1.82-21.99)	Fixed	PRAW	0.64	0.42	0

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M. xenopi	1	7.69 (1/13)	-	-	-	-	-
M. lentiflavum	1	5.26 (1/19)	-	-	-	-	-
M. shimoidei	1	3.03 (1/33)	-	-	-	-	-

Abbreviations: MAC: *Mycobacterium Avium Complex*; M. chelonae: Mycobacterium chelonae; M. kansasii: Mycobacterium kansasii; M. abscessus: Mycobacterium abscessus; M. fortuitum: Mycobacterium fortuitum; M. gordonae: Mycobacterium gordonae; M. shimoidei: Mycobacterium shimoidei; M. avium intracellulare: Mycobacterium avium intracellulare; M. lentiflavum: Mycobacterium lentiflavum; M. simiae: Mycobacterium simiae; M. xenopi: Mycobacterium xenopi; M. peregrinum: Mycobacterium peregrinum; PRAW=Raw data; PLOGIT=Logit transformation; PFT=Freeman-Tukey double arcsine transformation.

Notes: Proportion in each study = number of specimens tested positive for a specific subspecies/total NTM-positive specimen in the study.

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Figure Legends:

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram. Our initial analysis of the 15 studies selected indicates that 4 from bronchiectasis medical/registry/referral centers exaggerated the prevalence, causing an overestimation of the prevalence of NTM in adults with non-CF bronchiectasis. Thus, the 4 papers were excluded, and 11 studies were included for the meta-analysis.

Figure 2. Forest plot of all included fifteen studies.

Figure 3. Forest plot of a sensitivity analysis after four outlier studies were excluded.

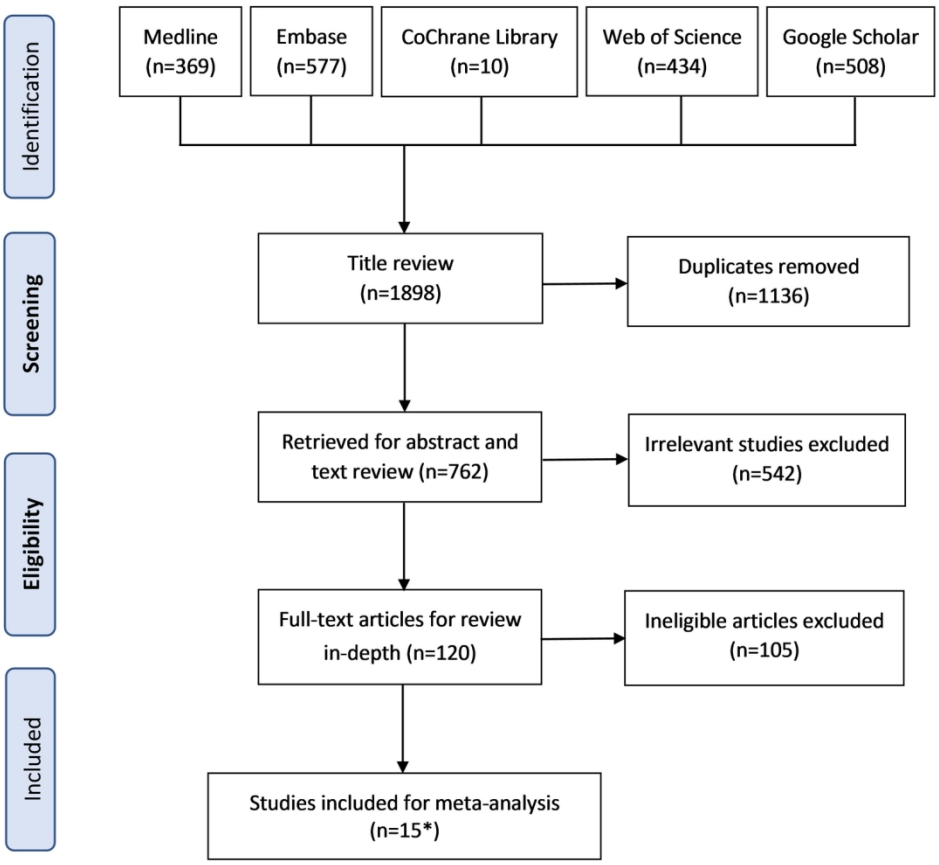
Figure 4. Forest plot for subgroup analysis on sample size.

Figure 5. Forest plot for subgroup analysis by geographical location.

Figure 6. Forest plot for subgroup analysis by study type (Retrospective vs. Prospective).

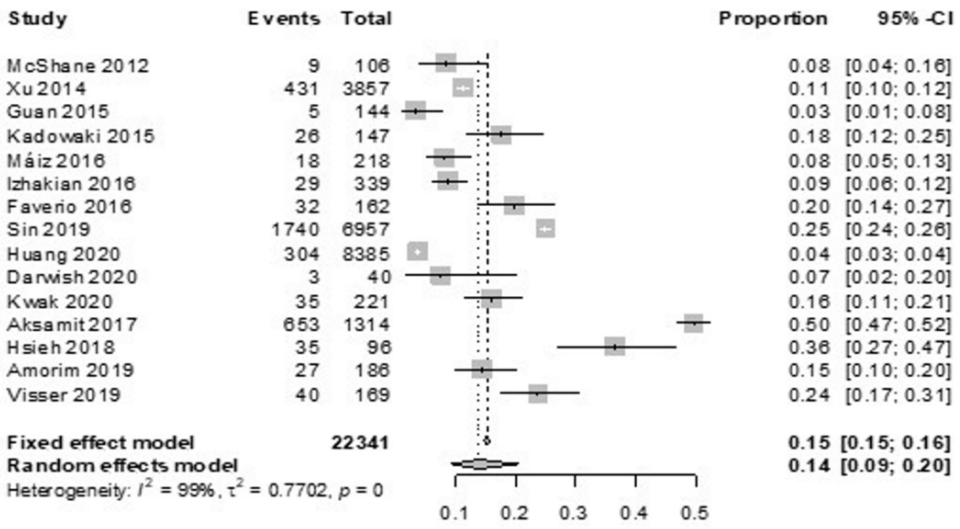
Figure 7. Forest plot for subgroup analysis on the methods of specimen examination.

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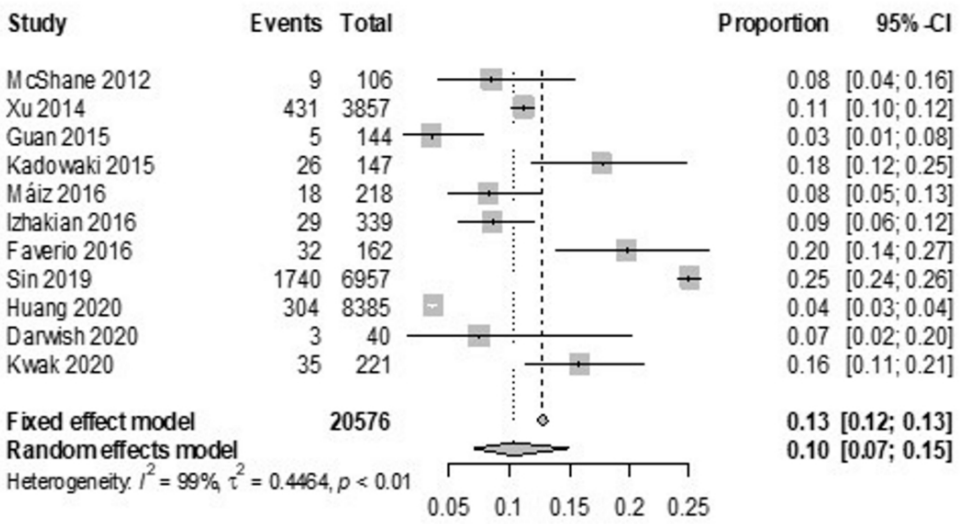
Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram. Our initial analysis of the 15 studies selected indicates that 4 from bronchiectasis medical/registry/referral centers exaggerated the prevalence, causing an overestimation of the prevalence of NTM in adults with non-CF bronchiectasis. Thus, the 4 papers were excluded, and 11 studies were included for the meta-analysis.

144x127mm (300 x 300 DPI)



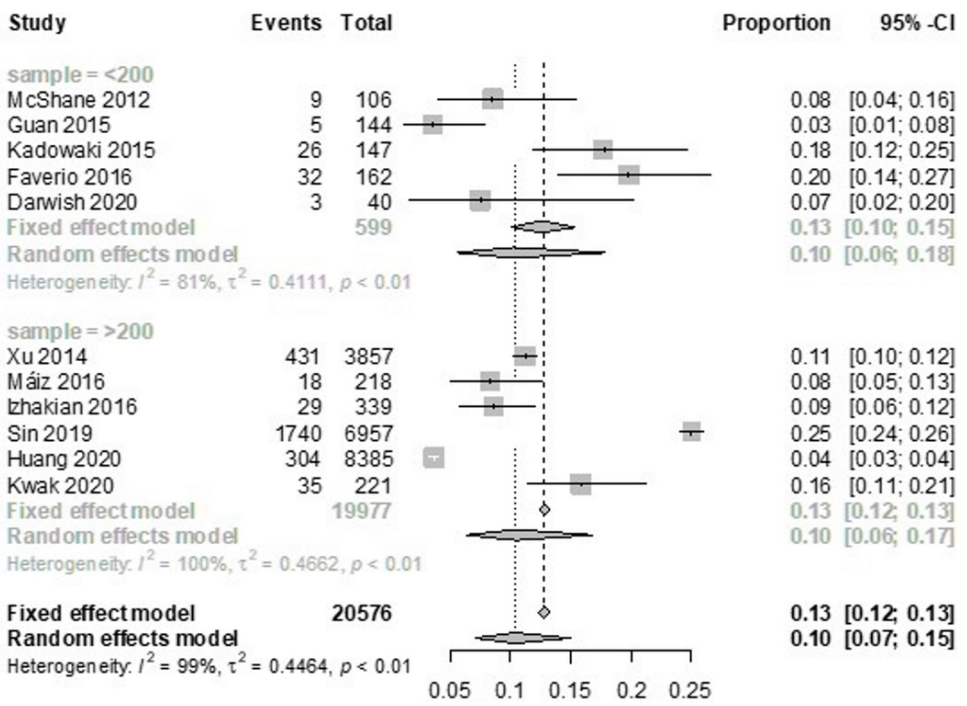
Forest plot of all included fifteen studies.

92x52mm (300 x 300 DPI)



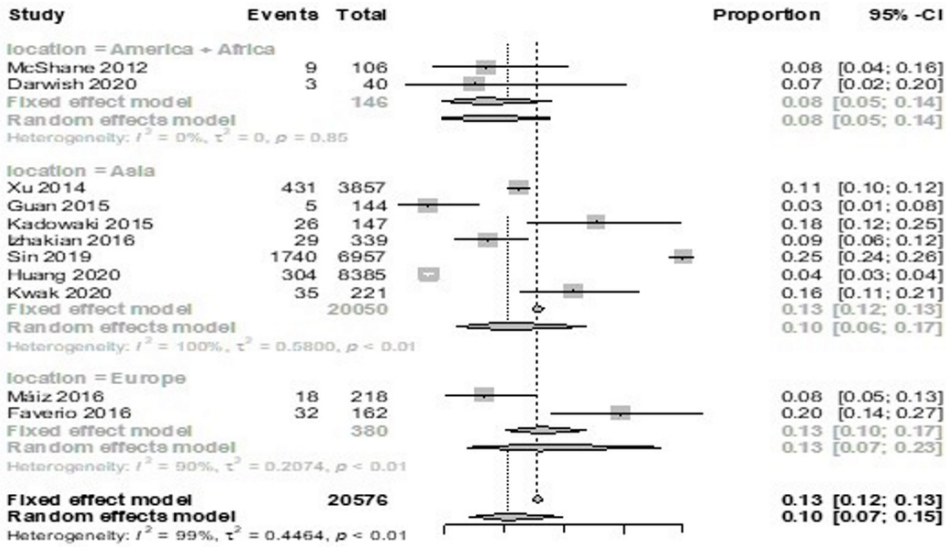
Forest plot of a sensitivity analysis after four outlier studies were excluded.

92x50mm (300 x 300 DPI)



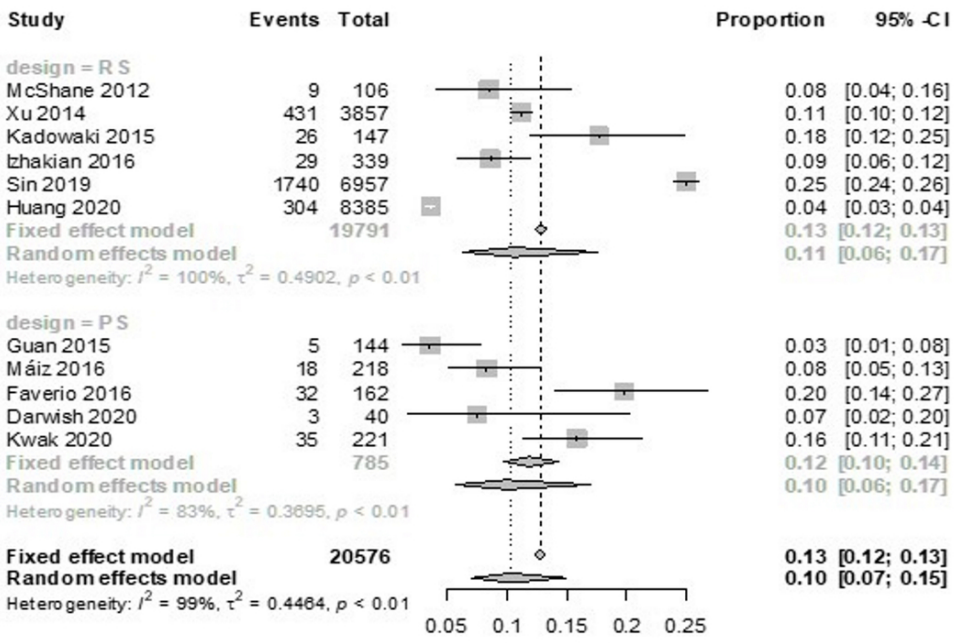
Forest plot for subgroup analysis on sample size.

92x66mm (300 x 300 DPI)



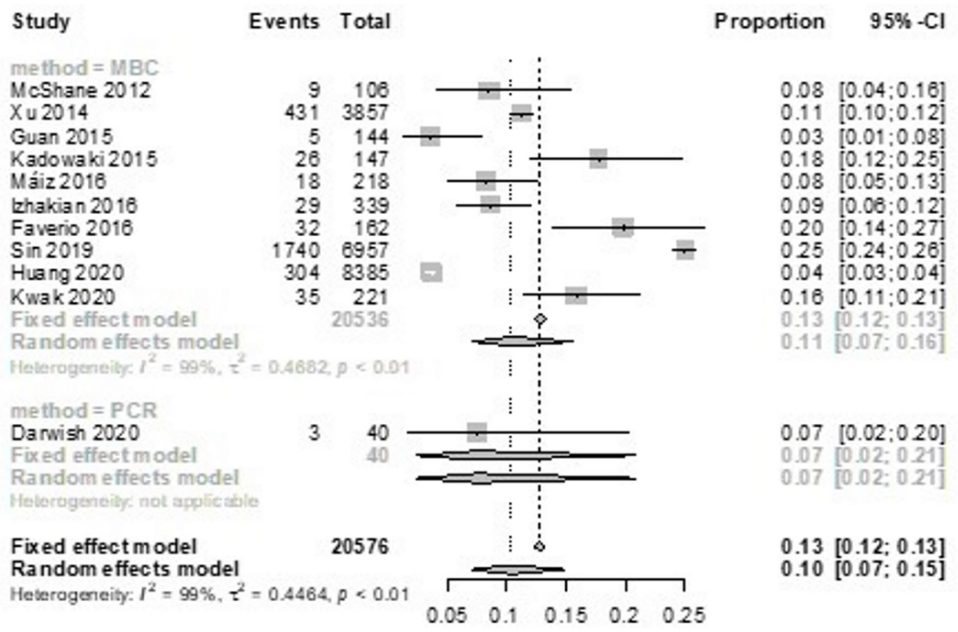
Forest plot for subgroup analysis by geographical location.

92x50mm (300 x 300 DPI)



Forest plot for subgroup analysis by study type (Retrospective vs. Prospective).

92x59mm (300 x 300 DPI)



Forest plot for subgroup analysis on the methods of specimen examination.

92x59mm (300 x 300 DPI)

Online Supplementary File 1: PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	3-4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	5-6
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	6-7
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	7-8
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	4, 7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	7-8
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	9
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	11-12
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	7-8
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	8, 10
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	9, 14
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	14
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	9
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	8
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	9

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Online Supplementary File 1: PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	10
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	8, 13
Study characteristics	17	Cite each included study and present its characteristics.	11-12
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table S1
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	13-14
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	10-18
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	10-18
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	10-18
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	10-18
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	17
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	10-18
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	19-20
	23b	Discuss any limitations of the evidence included in the review.	21
	23c	Discuss any limitations of the review processes used.	21-22
	23d	Discuss implications of the results for practice, policy, and future research.	20-21
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	4
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	4
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	22
Competing interests	26	Declare any competing interests of review authors.	22
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	22

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71



Online Supplementary File 1: PRISMA 2020 Checklist

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Global prevalence of nontuberculous mycobacteria in adults with non-cystic fibrosis bronchiectasis 2006-2021: a systematic review and meta-analysis

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Global prevalence of nontuberculous mycobacteria in adults with non-cystic fibrosis bronchiectasis 2006-2021: a systematic review and meta-analysis

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ABSTRACT

Objective To accurately estimate the global prevalence of NTM in adults with non-CF bronchiectasis and determine the proportion of NTM species and subspecies in clinical patients from 2006 to 2021.

Design Systematic review and meta-analysis using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.

Data sources Medline, Embase, Cochrane Library, and Web of Science were searched for articles published between 2006 and 2021.

Eligibility criteria for selecting studies We included all the prospective or retrospective studies without language restrictions and all patients were adults (≥18 years of age) with non-CF bronchiectasis. The studies estimated the effect size of the prevalence of NTM with a sample size ≥40, and patients were registered in and after 2006.

Data extraction and synthesis Two reviewers screened the titles, abstracts and full texts independently. Relevant information was extracted and curated into tables. Risk of bias was evaluated following the Cochrane Collaboration's tool. Meta-analysis was performed with software R Statistics 3.6.3 using random effect model with 95% confidence interval. I^2 index and Q statistics were calculated to assess the heterogeneity, and mixed-effects meta-regression analyses were performed to identify the sources of heterogeneity. The proportions of NTM subspecies were examined using Shapiro-Wilk normality test in R.

Results Of all the 2014 studies yielded, 24 met the inclusion criteria. Of these, 14 were

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identified to be randomized controlled studies and included for an accurate estimation. The global prevalence of NTM in adults with non-CF bronchiectasis from 2006 to 2021 was estimated to be approximately 10%, with great variations primarily due to geographical location. Mycobacterium avium complex (MAC) was the most common subspecies, followed by *M. simiae* and *M. gordonae*.

Conclusions The prevalence of NTM in adults with non-CF bronchiectasis has been on the rise and the most common subspecies changed greatly in recent years. More cohort studies should be done in many countries and regions for future estimates.

PROSPERO registration number CRD42020168473

Strengths and limitations of this study

- This study systematically reviewed the data over the past 16 years according to the PRISMA guidelines and employed the Cochrane Collaboration's tool to evaluate risk of bias.
- Factors causing overestimation, such as clinical data from bronchiectasis medical/ referral/registry centers, were considered during the metal-analysis.
- The significance of NTM subspecies in clinical practice was examined.
- Due to strict selection criteria, much data had to be excluded in this study, resulting in limited data from Africa and America.
- For molecular method (PCR), only one study was eligible with limited sample size,

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making a comparison of detection methods impossible.

Key words: non-cystic fibrosis bronchiectasis, nontuberculous mycobacteria, prevalence, meta-analysis, subspecies.

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INTRODUCTION

The prevalence of non-cystic fibrosis (CF) bronchiectasis increased dramatically in recent years, and it is the third most common respiratory disease after asthma and chronic obstructive pulmonary disease (COPD) [1–3]. The airways of patients with non-CF bronchiectasis are chronically colonized by a variety of pathologic microorganisms, such as yeasts, filamentous fungi, and nontuberculous mycobacteria (NTM) [4]. Currently it is widely believed that non-CF bronchiectasis and NTM infection are interrelated [5], and NTM causes human disease as opportunistic pathogens within a complex clinical context [6,7], such as COPD or CF [8]. Several factors, including age, sex, cigarette smoking, HIV infection, and underlying health conditions are associated with the susceptibility to NTM infection [9–11]. People living with HIV is particularly vulnerable to NTM infection due to immunodeficiency conditions [9]. Host factors, such as smoking, are believed to be important determinants of the susceptibility to NTM infection [10]. The age for the highest prevalence of NTM pulmonary disease was in the 50s for women and in the 70s for men, except for those over 80 years of age [12]; whereas NTM is rarely isolated in children and adolescence with Non-CF bronchiectasis [13], especially in those under 15 years old [14]. Thus, this study focused exclusively on the prevalence of NTM in adults. Among these factors, concomitant bronchiectasis may be the strongest factor associated with NTM infection [15]. As a result, although the mechanism of NTM infection, as well as its impact on clinical outcomes, is not well understood [16], it is hypothesized that NTM infection may lead to bronchiectasis [5].

The reported prevalence of NTM in the population with non-CF bronchiectasis varied widely among studies [11,17–19], and the NTM prevalence in adults with non-CF bronchiectasis from some studies might be overestimated as the data originated from

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bronchiectasis medical/referral/registry centers, where NTM in respiratory secretions were routinely screened [11]. Meanwhile, studies worldwide have discovered great geographic variation among species of NTM [20,21], which may also account for the variability in the prevalence of NTM pulmonary disease. Geographic factors, such as climate, geological distribution, and regional differences, may affect NTM activity, but the underlying mechanism remains elusive [22]. Moreover, sample size [23], detection methods [24], and study design [25], may also contribute to the variations. In addition, *M. kansasii*, *M. szulgai*, and *M. malmoense* have been recognized as causative agents in most patients, whereas *M. gordonae, M. terrae,* and *M. fortuitum* were less virulent and they were usually believed to be contaminants rather than causative agents [26,27]. Therefore, determining the prevalence of NTM subspecies in patients with non-CF bronchiectasis is important to understand the role of NTM in the disease, and may guide clinical treatments by dealing with the microbes in an earlier stage and improving patient outcomes.

Multiple studies have found that NTM isolation rates are increasing in recent years, which may be attributable to the advancement of detection methods [28–30]. Chu et al. reported that the prevalence of NTM was 9.3% in 2014 [23], and *M. abscessus* and MAC were the two most prevalent NTM subspecies. However, all the clinical data in this study was collected prior to 2006, which could not reflect the recent trends. It is therefore necessary to update the estimate of the global prevalence of NTM after 2006 for better understanding of the pathogenesis of non-CF bronchiectasis. Herein we aim to determine the global prevalence of NTM in adults with non-CF bronchiectasis from 2006 to 2021, explore the possible source of heterogeneity in the prevalence of NTM pulmonary disease,

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and identify the significance of NTM subspecies in clinical practice.

METHODS

We conducted a systematic review on NTM airway colonization in adults with non-CF bronchiectasis from 2006 to 2021 according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (PRISMA-P checklist 2020; **Online Supplementary File 1**) [31], and performed a meta-analysis on the global prevalence of NTM, as well as the proportion of their subspecies.

Search strategy and selection criteria

We searched the databases on Medline, Embase, Cochrane Library, and Web of Science, for articles published from 2006 to December 2021, using keywords “nontuberculous mycobacter*”, “NTM”, “non-tuberculous mycobacter*”, “nontuberculous mycobacter*”, “atypical Mycobacter*”, “bronchiectasis”, “sputum”, “microbiome”, “bronchiectas*”, and “Kartagener*”. Databases were searched without language restrictions (**Supplementary Table S1)**. We also reviewed the references cited by selected articles to identify additional studies meeting the inclusion criteria. Two researchers (YCZ and WM) searched titles and abstracts and selected articles based on the inclusion criteria; any discrepancies were solved by consensus with the help of a third reviewer (SWW).

We included all observational studies fulfilling the following criteria: 1) prospective or retrospective studies; 2) studies in adults (≥18 years of age) with non-CF bronchiectasis as defined by the authors; 3) studies estimating the effect size of the prevalence of NTM; 4) sample size ≥40; 5) clinical patients or the vast majority of clinical patients in and after 2006; and 6) publications before 2021. We excluded studies if: 1) it only described NTM data unassociated with bronchiectasis; 2) duplicate studies or records or did not calculate

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the prevalence of NTM in the patients of bronchiectasis; 3) participants included cystic fibrosis; 4) cultures included both NTM and tuberculous mycobacteria simultaneously; or, 5) case reports, commentary, or review articles.

Data extraction

Data was extracted by two reviewers (YCZ and WM) independently using the standard protocol and discrepancies were resolved through consultation with the third researcher (SWW). The following information of included studies were extracted: 1) last name of the first author, year of publication, country of the population studied; 2) time of study; 3) sample size; 4) mean age of patients; 5) data source; 6) cohort study design, whether retrospective or prospective; 7) specimen source such as sputum or bronchoscopy specimen; 8) laboratory method of NTM tested; 9) definition of NTM positive; and, 10) infection rate of NTM in adults with non-CF bronchiectasis with the classification of NTM. For missing information, we communicated with the authors; if the information was unavailable after communications, this study was excluded. Two reviewers (YCZ and WM) evaluated the risk of bias in individual studies following the Cochrane Collaboration's tool [32].

Quantitative analysis

Software R Statistics version 3.6.3 was used for statistical analysis. The random effect model with a 95% confidence interval (CI) was adopted as we included observational studies and presumed high study heterogeneity. The I^2 index and Q statistic were calculated to assess study heterogeneity. The original prevalence of patients positive for NTM was tested for normality using the Shapiro-Wilk normality test in R. The results of the test determine if the original prevalence needs transformation before being pooled, and risk of bias assessment determines if a study should be excluded for sensitivity

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analysis. Mixed-effects meta-regression analyses were performed to identify possible sources of heterogeneity. Proposed moderator variables include the method of NTM specimen examination (MBC versus PCR), study design (prospective versus retrospective), sample size (both as a categorical variable and a continuous variable) and geographic location of the participants. A p-value less than 0.1 was defined as statistically significant. Subgroup analyses was conducted by stratifying studies by proved moderator variable based on the results of meta-regression analyses. Lastly, the prevalence of each NTM subspecies reported in the finally included studies was examined.

Patient and public involvement

There was no direct patient or public involvement in this review.

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RESULTS

1. Characteristics of eligible studies

The systematic review yielded 2014 articles. After eliminating duplicates and articles that did not match the inclusion criteria, a total of 24 independent studies including 26,944 patients with non-CF bronchiectasis were included in this systematic review and meta-analysis (**Figure 1**). The characteristics of the 24 studies were summarized (**Supplementary Table S2)**.

Figure 1 Here

2. Overall effects

2.1 Initial analysis of all the selected studies

The results of the Shapiro-Wilk normality test showed that the prevalence requires transformation before pooling because they do not follow a normal distribution (W = 0.857, p-value = 0.003). The Freeman-Tukey double arcsine transformation of the original prevalence was then made and pooled. Pooled analysis of the 24 included studies shows that the prevalence of NTM in adults with non-CF bronchiectasis was estimated at 13.76% (95% CI [8.53%-19.97%]) (**Figure 2**). Result of the Chi-squared test for heterogeneity was 99.5% for the 24 studies, indicating a great degree of variations among these studies.

Figure 2 Here

2.2 Overestimation existed in the selected studies

Figure 2 indicated that the clinical data from 4 studies (Park 2016 [33], Aksamit 2017 [17], Hsieh 2018 [34], and Choate 2021 [35]) was visually identified as outliers from the forest plot, whose prevalence (50%, 36%, 24%, and 48%) was substantially higher and the confidence intervals had no or little overlaps with the CIs of other studies or the total effect size, indicating studies from bronchiectasis medical/registry/referral centers

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tremendously increased the estimated prevalence of NTM. Similarly, the clinical data from the other 4 studies (Amorim 2019 [36], Visser 2019 [29], Dhar, 2019 [37], Metersky 2021 [38]) was also from bronchiectasis referral centers. Shteinberg and Aksamit believed that the estimates of NTM prevalence from bronchiectasis referral centers were exaggerated [39,40], suggesting that including these 8 studies would inevitably overestimate the prevalence of NTM in adults with non-CF bronchiectasis and therefore, should be excluded. In another study (Máiz, 2016), only the patients who had been followed for at least 5 years were included in analysis [18]. Additionally, Kwak's study in 2020 only included the patients who had participated in a non-NTM bronchiectasis cohort and then studied the NTM infection afterwards [41]. The patients in both studies were not randomly selected, and that may also result in inaccurate estimates. Thus, our risk of bias assessment (**Supplementary Table S3**) suggested that, out of all the 24 studies selected, 10 studies should be excluded from the sensitivity analysis to avoid an overestimation of the prevalence of NTM.

2.3 An accurate estimate of the prevalence of NTM infection from 2006 to 2021

Based on the analysis above, 14 independent studies were included in the meta-analysis for an accurate estimate of the prevalence of NTM in adults with non-CF bronchiectasis (**Table 1**). These studies encompassed 21,056 patients with non-CF bronchiectasis and NTM has been isolated from 2643 patients. All the patients were ≥18 years old. The prevalence of NTM in adults with non-CF bronchiectasis ranged from 1.0% to 25%. The results of the Shapiro-Wilk normality test showed that the prevalence follows a normal distribution (W = 0.877, p-value = 0.053) and needs no transformation. Our sensitivity analysis showed that the prevalence of NTM infection in adults with non-CF bronchiectasis from 2006 to 2021 was 9.75% (95% CI [5.41%-14.09%]) (**Figure 3**).

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Results of the Chi-squared test for heterogeneity were 99.3%, suggesting a great degree of variations among the 14 included studies. Notably, this was an overall estimate of the prevalence of NTM infection in adults with non-CF bronchiectasis, regardless of underlying health condition such as HIV, age, sex, and smoking.

Figure 3 Here

Table 1: Characteristics of studies included in the systematic review and meta-analysis

Author & year	Country	Time of sampling	Mean age (Y)	Sample size	Patients with NTM	Study design	Method of detection	NTM [%]
1. McShane, et al. [42] 2012	USA	2009-2011	≥18	106	9	RS	Sputum culture	8.5
2. Xu, et al. [43] 2014	China	2009-2012	47.4	3857	431	RS	Sputum culture	11.2
3. Guan, et al. [44] 2015	China	2012-2013	44.6	144	5	PS	Sputum culture	3.5
4. Kadowaki, et al. [45] 2015	Japan	2008-2012	73	147	26	RS	Sputum culture	17.7
5. Izhakian, et al. [46] 2016	Israel	2006-2014	64	339	29	RS	Bronchoalveolar /lavage cultures	8.6
6. Faverio, et al. [47] 2016	Italy	2006-2014	65	162	32	PS	Bronchoalveolar /sputum culture	19.8
7. Buscot, et al. [48] 2016	France	2002-2012	61.0	196	7	RS	Sputum/ bronchoalveolar lavage culture	3.6
8. Dimakou, et al. [49] 2016	Greece	2009-2014	60.5	205	2	PS	Sputum culture	1.0
9. Sin, et al. [19] 2019	Korea	2005-2016	59.6	6957	1740	RS	Sputum culture	25
10. Pieters, et al. [50] 2019	Netherla nd	2012-2016	60	120	6	RS	Sputum culture	5.0
11. Huang, et al. [51] 2020	China	2002-2016	65.5	8385	304	RS	Sputum culture	3.6
12. Darwish, et al. [52] 2020	Egypt;	2017-2018	55.2	40	3	PS	Sputum PCR	7.5
13. Sharif, et al. [53] 2020	Pakistan	2017-2019	NA	196	2	PS	Sputum culture	1.0
14.Yin, et al. [54] 2021	China	2018-2020	62	202	47	RS	Sputum/BAL culture	23.3

Abbreviations: NA: Not Available; NTM: Nontuberculous Mycobacteria; MBC: mycobacteria culture; AFB: acid fast bacillus smear; BAL: bronchoalveolar lavage; RS: Retrospective Study; PS: prospective Study; UK: United Kingdom; US: United States. PCR: Polymerase Chain Reaction.

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3. Meta-regression and subgroup analysis

To understand the mechanisms of variations, we sub-grouped the included 14 studies based on 4 characteristics (**Table 2**). We performed mixed-methods meta-regression analyses using Metafor package in R. Moderating variables including the method of NTM specimen examination (R2 analog=0.00%, test of moderators *p*=0.802), sample size (as continuous variable, R2 analog=0.00%, test of moderators *p*=0.479; as categorical variable, R2 analog=0.00%, test of moderators p=0.390) and study design (R2 analog=4.82%, test of moderators *p*=0.222) failed to explain the heterogeneity among studies. The moderator geographic location of the participants explained 14.2% of the between-study variance (test of moderators *p*=0.083).

Subsequently, we conducted a subgroup analysis by stratifying studies into reports from East Asian (including China, Korea and Japan) and other geographical locations. The pooled NTM prevalence of the six East Asian studies was 7.50% higher than that of the eight studies from other countries (13.87%, 95% CI [6.23%-21.52%] vs. 6.37%, 95% CI [2.38-10.36%]) (**Figure 4**). The Chi-squared test found better homogeneity among studies from other countries. These analyses suggested that the variations among the 14 included studies were not due to sample sizes, study types, and specimen examination methods, but geographic locations. Further studies on regional differences, such as climate, soil, water, culture, may help to understand the epidemiology of NTM infection in adults with non-CF bronchiectasis.

Figure 4 Here

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Table 2: Subgroup classification for data analysis

Study	Time of clinical data	Number of NTM positive	Number of patients	Sample size	Study location	Study type	Culture method
1. McShane, 2012	2009-2011	10	106	< 200	Others	RS	MBC
2. Xu, 2014	2009-2012	431	3857	≥200	East Asia	RS	MBC
3. Guan, 2015	2012-2013	5	144	< 200	East Asia	PS	MBC
4. Kadowaki, 2015	2008-2012	26	147	< 200	East Asia	RS	MBC
5. Izhakian, 2016	2006-2014	29	339	≥200	Others	RS	MBC
6. Faverio, 2016	2006-2014	32	162	< 200	Others	PS	MBC
7. Buscot, 2016	2002-2012	7	196	< 200	Others	RS	MBC
8. Dimakou, 2016	2009-2014	2	205	≥200	Others	PS	MBC
9. Sin, 2019	2005-2016	1740	6957	≥200	East Asia	RS	MBC
10. Pieters, 2019	2012-2016	6	120	< 200	Others	RS	MBC
11. Huang, 2020	2002-2016	304	8385	≥200	East Asia	RS	MBC
12. Darwish, 2020	2017-2018	3	40	< 200	Others	PS	PCR
13. Sharif, 2020	2017-2019	2	196	< 200	Others	PS	MBC
14.Yin, 2021	2018-2020	47	202	≥200	East Asia	RS	MBC
Sum		2,644	21,056				

Abbreviations: RS: Retrospective Study; PS: Prospective Study; MBC: Mycobacteria Culture; AFB: Acid-fast Bacillus smear; PCR: Polymerase Chain Reaction.

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4. Subspecies analysis

Lastly, we estimated the prevalence of NTM subspecies in non-CF adult bronchiectasis patients reported in the included 14 studies. Because more than one NTM subspecies might be identified in a specimen, we used the times of a subspecies being identified for pooling or calculating the proportion (**Supplementary Table S4**). The method of data transformation was determined considering the characteristics of the raw proportions. Out of all 14 studies, nine reported the number of confirmed cases or specimens of each NTM subspecies (**Table 3**). Our results showed that the most widely identified NTM species in our review is MAC, reported in nine studies with a pooled proportion of 77.6% (95% CI [64.0%-91.1%]), followed by *M. simiae* (17.28%, 95% CI [0.00%-36.80%]) and *M. gordonae* (10.50%, 95% CI [1.65%-19.3%]).

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Table 3: The proportion of subspecies in NTM positive specimens

NTM type	No. of studies	Proportion (95% CI [%])	Effect model	Method of data transformation	Test of heterogeneity		
					Q	P-value	I^2 [%]
MAC	9	77.6 (64.0-91.1)	Random	PRAW	66.0	<0.001	87.9
M. abscessus	4	7.83 (0.00-15.8)	Random	PRAW	10.47	0.015	71.4
M. chelonae	5	5.70 (2.21-9.18)	Fixed	PRAW	0.42	0.981	0
M. gordonae	2	10.5 (1.65-19.3)	Fixed	PRAW	0.23	0.635	0
M. fortuitum	3	5.27 (0.64-9.91)	Fixed	PRAW	0.35	0.841	0
M. simiae	2	17.28(0.00-36.80)	Random	PRAW	3.20	0.073	68.8
M. kansasii	3	4.70 (0.41-8.98)	Fixed	PRAW	0.70	0.705	0
Undetermined	2	2.31(0.00-6.43)	Fixed	PRAW	1.31	0.253	23.5
M. xenopi	1	7.69 (1/13)	-	-	-	-	-
M. shimoidei	1	3.03 (1/33)	-	-	-	-	-
M. terrae	1	2.1%(1/47)	-	-	-	-	-

Abbreviations: MAC: *Mycobacterium Avium Complex*; *M. abscessus*: Mycobacterium abscessus; *M. chelonae*: Mycobacterium chelonae; *M. gordonae*: Mycobacterium gordonae; *M. fortuitum*: Mycobacterium fortuitum; *M. simiae*: Mycobacterium simiae; *M. kansasii:* Mycobacterium kansasii; *M. xenopi*: Mycobacterium xenopi; *M. shimoidei*: Mycobacterium shimoidei; *M. terrae:* Mycobacterium *terrae*; PRAW=Raw data.

Note: Proportion in each study = number of specimens tested positive for a specific subspecies/total NTM-positive specimen in the study; the proportions of NTM subspecies identified in only one study were calculated, rather than pooled.

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DISCUSSION

Main Findings

Our systematic review including 24 studies from multiple geographic locations in the initial meta-analysis, found that the global prevalence of NTM in adults with non-CF bronchiectasis from 2006 to 2021 was 14%. However, when the 10 studies with source bias in the clinical data were excluded from meta-analysis, this percentage decreased to 10%. An updated, accurate estimate of global prevalence of NTM is of clinical significance, and data with source of bias should be excluded in such an estimation [5,16]. As a result, with available data, the most accurate estimation of global prevalence of NTM infection in adults with non-CF bronchiectasis should be 10%.

Our subgroup analyses found that of all the 4 factors we investigated, the great variation of the prevalence of NTM among studies was primarily due to geographic difference, whereas sample size, study type, and detection method only slightly affected the estimates. The pooled NTM prevalence of East Asia (13.87%, 95% CI [6.23%-21.52%]) was 7.50% higher than that of other continents on the Earth (6.37%, 95% CI [2.38%-10.36%]) (**Figure 4**). Interestingly, the prevalence of NTM among countries or within the same continent, or even among regions within the same country also varied greatly (**Figure 3**). The mechanisms for this geographic variation remain elusive, and unknown factors may contribute to the high prevalence of NTM in some countries or regions. It was speculated that environment factor such as climate, soil, and water [6–8,55], and regional differences in techniques [56] may account for the geographic variations of NTM prevalence. Further studies on the geographic variations may contribute to the prevention of NTM in adults with non-CF bronchiectasis.

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Comparison with literature

Compared to Chu's study in 2014, of which all the clinical data were between 1990 and 2006 [23], the clinical data in this study were primarily from 2006 to 2021. Thus, this study provided an updated estimate of the NTM prevalence in adults with non-cystic fibrosis bronchiectasis. Moreover, within the same amount of time, the number of studies tripled from 8 to 24, indicating an increasing interest in the prevalence of NTM in patients with non-CF bronchiectasis during the past 15 years [30]. Without taking the source bias into consideration, Chu's study in 2014 estimated that the prevalence of NTM from 1990 to 2006 was 9.3%. In contrast, our initial analysis estimated that the prevalence of NTM from 1990 to 2006 was 13.76%, suggesting an increasing trend of NTM infections in patients with non-CF bronchiectasis.

After an in-depth analysis of the report in 2014 by Chu et al [23], we found that the prevalence of NTM from 1990 to 2006 might have been overestimated, primarily because out of all 8 studies included in this study, two were from bronchiectasis medical/referral centers (Koh et al., 2005; Tabarsi et al., 2009) [57,58]. Koh et al. reported a prevalence of NTM as high as 30.0%. However, many patients in this study were not bronchiectasis but bronchiolitis [59]. Tabarsi, et al. reported that the prevalence of NTM was 15.2%, but the cases might have included multidrug-resistant tuberculosis [60]. Thus, we believe that in this study, only 6 studies were valid for the estimate of the prevalence of NTM in patients with bronchiectasis. Accordingly, we re-calculated the prevalence of NTM in patients with bronchiectasis of the 6 valid studies, finding that the real prevalence of NTM from 1990 to 2006 was not 9.3%, but 5% (**Figure S1)**. This systematic review and meta-analysis estimated that the global prevalence of NTM from 2006 to 2021 was

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approximately 10% (9.75%, 95% CI [5.41%-14.09%]), implicating the increasing trend in the prevalence of NTM infections in adults with non-CF bronchiectasis over the past 15 years.

Moreover, Chu, et al. reported that from 1990 to 2006, the three most prevalent NTM subspecies were *M. abscessus* (43.2%, 95% CI [26.6%~60.9%]), MAC (25.7%, 95% CI [12.6%-38.8%]), and *M. kansasii* (3.7%, 95% CI [0.1%~7.3%]). This analysis discovered that from 2006 to 2021 the three most prevalent NTM subspecies turned to be MAC (77.6% 95% CI [64.0%-91.1%]), *M. simiae* (17.28%, 95% CI [0.00%~36.80%]), and *M. gordonae* (10.50%, 95% CI [1.65%~19.30%]). Evidently, MAC has turned to be the most common subspecies in the past 15 years. Studying the underlying mechanism may help to understand the role of NTM infection in the pathogenesis of non-CF bronchiectasis.

Implications for research and practice

The increasing interest in NTM infection in non-CF bronchiectasis worldwide in recent years indicates the microbes may play an important role in the pathogenesis of the disease [47]. In general, of all patients with a positive culture, only 25~60% meet the criteria for NTM pulmonary disease [26]. Our data showed that up to 65%, or even up to 90% of the total NTM infections in bronchiectasis patients were subspecies MAC, similar to the data previously estimated [59,61,62]. Some subspecies, such as *M. kansasii*, *M. szulgai*, and *M. malmoense,* were considered causative agents, whereas the others, such as *M. gordonae, M. terrae,* and *M. fortuitum* were believed to be less virulent contaminants [26,27]. Therefore, the 3 most prevalent NTM species converted from *M. abscessus*, MAC, and *M. kansasii* into MAC, *M. simiae*, and *M. gordonae*. This conversion might be clinically significant to reveal the role of NTM in the disease, thereby

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guiding clinical treatments and making an accurate prognosis [27,63]. However, a diagnosis of infection in lung disease does not dictate the initiation of antibiotic therapy against NTM species. Instead, the healthcare team should consider the potential risks and benefits for individual patients of a prolonged course of treatment with multiple antibiotics [64].

Strengths and limitations

In the study by Chu et al. in 2014, all the clinical data was prior to 2006. In contrast, the clinical data in this paper was between 2006 and 2021. There were no data and time overlaps between the two studies and thus, results from this study may represent the current trend of NTM prevalence in adults with non-CF bronchiectasis. Moreover, this study took source bias into consideration, thereby providing an updated and more accurate estimate of the prevalence of NTM in adults with non-CF bronchiectasis, which may be significant to direct clinical treatments.

This study has several limitations as well. First, only one study from Africa and one study from North America met the inclusion criteria, indicating that the data for these continents was less representative. Second, sample size is limited so in the analysis of determinants of NTM infections we have used $P < 0.1$ instead of conventional cut off $P < 0.05$ to define statistical significance. Third, the inspection methods may significantly affect detection rates [28–30]. After the inclusion criteria were applied, only one study used PCR to detect NTM species was available in the literature with limited sample size, making a comparison of different detection methods for species analysis impossible. Fourth, the American Thoracic Society/Infectious Disease Society of America (ATS/IDSA) Statement (2007) recommended that NTM positive culture results should be from at least

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two separate expectorated sputum samples[64]. Of the 14 studies included in this study, NTM detection method was described in details in 9 studies (including 1 using PCR), only 6 studies met the ATS/IDSA standard criteria, 5 did not report the number of specimens of NTM subspecies. The lab differences may partially account for the geographic variations of the NTM prevalence.

It is noteworthy to mention that the percentage presented herein (approximately 10%) is an overall estimate for the global prevalence of NTM in adults with non-CF bronchiectasis from 2006 to 2021. As per the prevalence of NTM in a specific country or region, retrospective and prospective studies are still needed for a customized and accurate estimate. Based on the analysis and issues presented in this study, other researchers may replenish the data and provide a more accurate and up-to-date estimation in the future.

CONCLUSIONS

Our systematic review and meta-analysis suggest that the global prevalence of NTM in adults with non-CF bronchiectasis from 2006 to 2021 is estimated to be approximately 10%, with major variations in the estimated prevalence primarily driven by geographical locations. MAC is the most common organism in non-CF bronchiectasis, followed by *M. simiae* and *M. gordonae*.

Abbreviations: NTM， nontuberculous mycobacteria； non-CF， non-cystic fibrosis； COPD, chronic obstructive pulmonary disease； CI, confidence interval； NA: Not Available; MBC: mycobacteria culture; AFB: acid fast bacillus smear; BAL: bronchoalveolar lavage; RS: retrospective study; PS: prospective study; UK: United

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Kingdom; US: United States; PCR: polymerase chain reaction.

Author contributions YZ, SP, and SWW contributed to the conception and design of the study. YZ, WM, SP, and SWW made substantial contributions to the acquisition of the data, data entry, or data analysis, and data interpretation. JZ provided daily assistance and professional comments on the manuscript. YZ, SP, and SWW were involved in drafting and revising the manuscript. All authors have approved the final manuscript submitted for publication.

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Online Supplementary File 1: PRISMA-P checklist; **Online Supplementary File 2:** 4 Supplementary tables and 1 Supplementary figure.

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64 Griffith DE, Aksamit T, Brown-Elliott BA, *et al.* An official ATS/IDSA statement: Diagnosis, treatment, and prevention of nontuberculous mycobacterial diseases. *Am J Respir Crit Care Med* 2007;**175**:367–416. doi:10.1164/rccm.200604-571ST

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Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram.

Figure 2. Forest plot of all included 24 studies from the systematic review.

Figure 3. Forest plot of a sensitivity analysis of 14 studies for an accurate estimate of global prevalence of NTM in adults with non-CF bronchiectasis 2006-2021.

Figure 4. Forest plot for subgroup analysis by geographical location.

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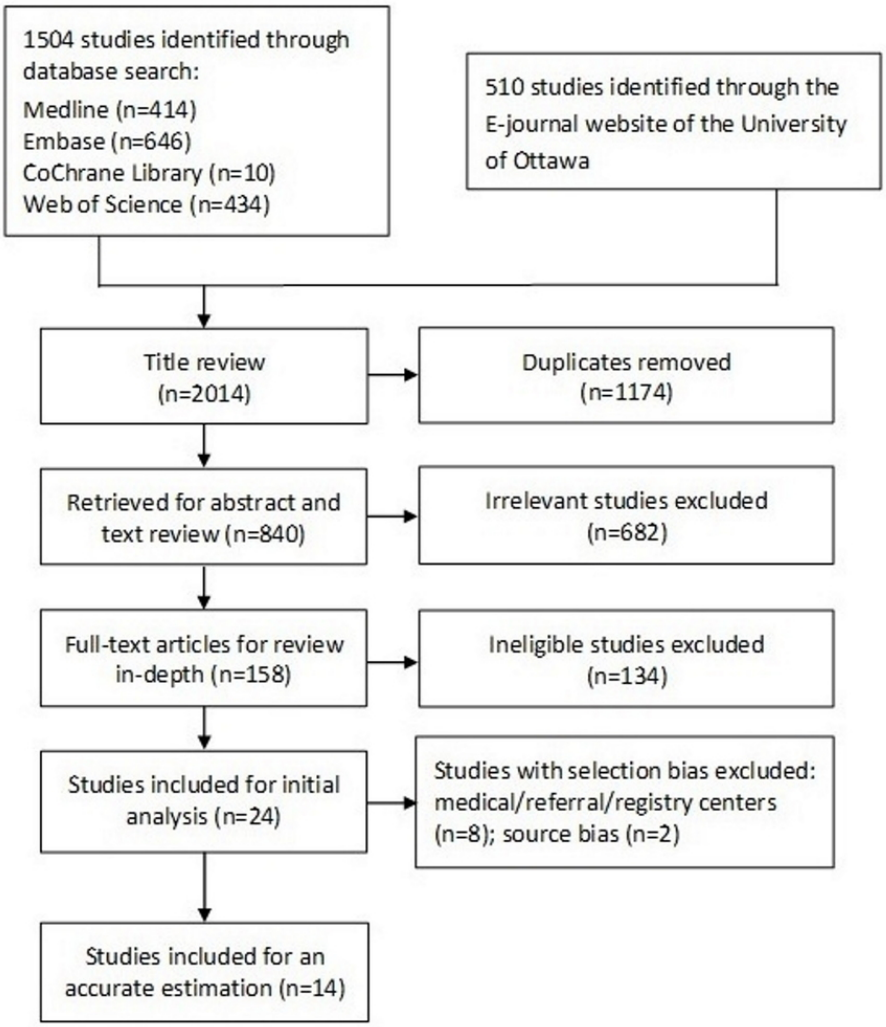


Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram.

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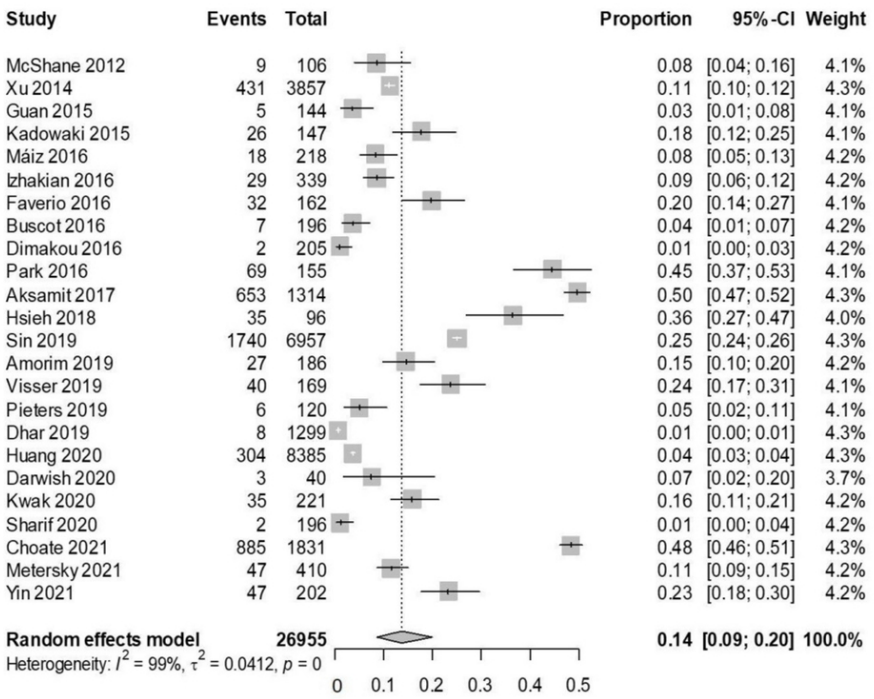


Figure 2. Forest plot of all included 24 studies from the systematic review.

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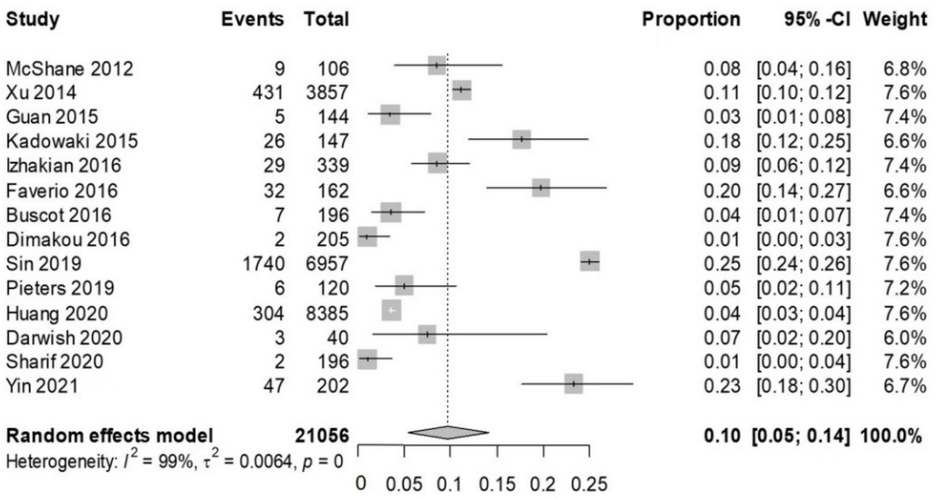


Figure 3. Forest plot of a sensitivity analysis of 14 studies for an accurate estimate of global prevalence of NTM in adults with non-CF bronchiectasis 2006-2021.

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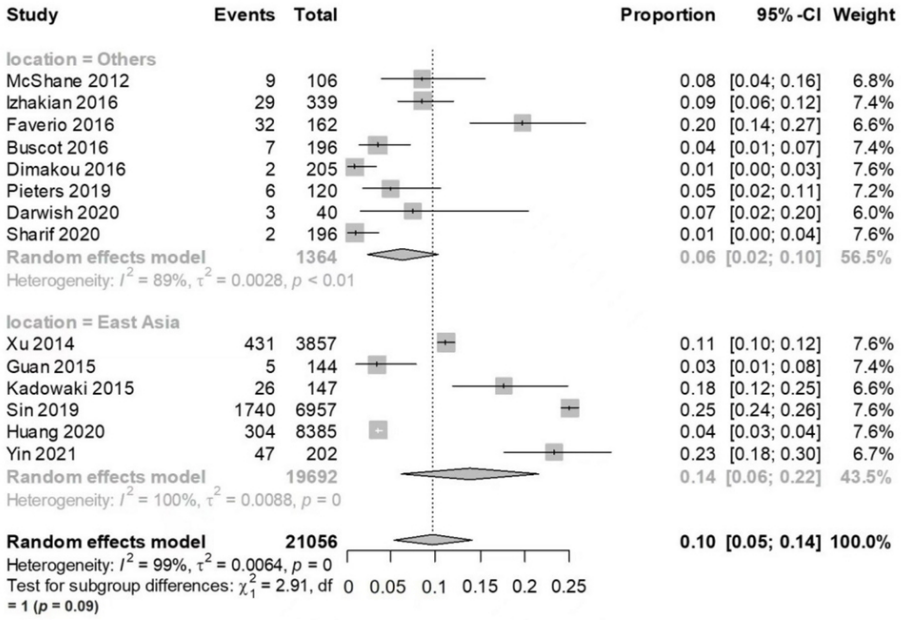


Figure 4. Forest plot for subgroup analysis by geographical location.

88x60mm (300 x 300 DPI)

Online Supplementary File 1: PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	4-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6-7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	2, 6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	6-7
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	7
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	7
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	7-8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	8
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	6-8
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	6-7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	7-8
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	8
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	7-8

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Online Supplementary File 1: PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	9
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	9-10
Study characteristics	17	Cite each included study and present its characteristics.	11
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table S2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	13
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	9-15
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	9-15
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	9-15
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	9-15
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	14
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	9-15
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	16-19
	23b	Discuss any limitations of the evidence included in the review.	19-20
	23c	Discuss any limitations of the review processes used.	19-20
	23d	Discuss implications of the results for practice, policy, and future research.	20
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	3
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	21
Competing interests	26	Declare any competing interests of review authors.	20
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	21

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Online Supplementary File 1: PRISMA 2020 Abstract Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesise results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review. (***At the end of the article***)	Yes
Registration	12	Provide the register name and registration number.	Yes

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71. For more information, visit: http://www.prisma-statement.org/

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Online Supplementary File 2

Supplementary files to 'Global prevalence of nontuberculous mycobacteria in adults with non-cystic fibrosis bronchiectasis 2006-2021: a systematic review and meta-analysis'.

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Table S1	Search strings	Page 2
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Figure S1	Re-calculating the prevalence of NTM in adults with non-cystic fibrosis bronchiectasis from 1990 to 2006 (Chu's study in 2014)	Page 9
References	……	Page 10

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Table S1: Search strings

Database and date	Yield	Duplicates	Search String used
Medline Jan 31, 2022	414	0	"Bronchiectasis" [Mesh]) AND ("Nontuberculous Mycobacteria" OR "non tuberculous mycobacter" [Mesh]) AND ("Atypical Mycobacter" [Mesh] OR ("Mycobacter abscessus" [Mesh]) OR "Mycobacter avium complex" [Mesh] OR "Mycobacter chelonae" [Mesh] OR "Mycobacter fortuitum" [Mesh] OR "Mycobacter kansasii" [Mesh] OR "Mycobacter marinum" [Mesh] OR "Mycobacter scrofulaceum" [Mesh] OR "Mycobacter smegmatis" [Mesh] OR "Mycobacter ulcerans" [Mesh] OR "Mycobacter xenopi" [Mesh])
Embase Jan 31, 2022	678	32	"Bronchiectasis" [Mesh]) AND ("Nontuberculous Mycobacteria" OR "non tuberculous mycobacter" [Mesh]) AND ("Atypical Mycobacter" [Mesh] OR ("Mycobacter abscessus" [Mesh]) OR "Mycobacter avium complex" [Mesh] OR "Mycobacter chelonae" [Mesh] OR "Mycobacter fortuitum" [Mesh] OR "Mycobacter kansasii" [Mesh] OR "Mycobacter marinum" [Mesh] OR "Mycobacter scrofulaceum" [Mesh] OR "Mycobacter smegmatis" [Mesh] OR "Mycobacter ulcerans" [Mesh] OR "Mycobacter xenopi" [Mesh])
Cochrane Library Jan 31, 2022	19	9	Bronchiectasis AND Nontuberculous Mycobacteria" OR "non tuberculous mycobacter
Web of Science Jan 31, 2022	504	70	"nontuberculous mycobacter" OR TOPIC: "non tuberculous mycobacter") OR TOPIC: "Atypical Mycobacter") OR TOPIC: (("Mycobacter abscessus" or "Mycobacter avium complex" OR "Mycobacter chelonae" OR "Mycobacter fortuitum" OR "Mycobacter* kansasii OR "Mycobacter marinum" OR "Mycobacter scrofulaceum" OR "Mycobacter smegmatis" OR "Mycobacter ulcerans" OR "Mycobacter xenopi")) OR TOPIC: "ntm"
Total	1615	111	
Other sources Jan 31, 2022	514		'Nontuberculous Mycobacteria' OR 'non-cystic fibrosis bronchiectasis'; publication date from 2006 to 2021, including citations
111 duplicates were removed to give a total of 2014 studies			

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Table S2: Characteristics of studies included in the initial analysis of selected studies

Author; year	Country	Data Source	Time of sampling	Mean age (Y)	Sample size	Patients with NTM	Study design	Method of detection	NTM [%]
1. McShane, et al. 2012 [1]	USA	University Referral Center	2009-2011	≥18	106	9	RS	Sputum culture	8.5
2. Xu, et al. 2014 [2]	China	Pulmonary Hospital	2009-2012	47.4	3857	431	RS	Sputum culture	11.2
3. Guan, et al. 2015 [3]	China	Outpatient Respiratory Clinics	2012-2013	44.6	144	5	PS	Sputum culture	3.5
4. Kadowaki, et al. 2015 [4]	Japan	Medical Center	2008-2012	73	147	26	RS	Sputum culture	17.7
5. Máiz, et al. 2016 [5]	**Spain**	**Teaching hospitals**	**2012-2015**	**55.7**	**218**	**18**	**PS**	**Sputum culture**	**8.3**
6. Izhakian, et al. [6] 2016	Israel	Rabin Medical Center	2006-2014	64	339	29	RS	Bronchoalveolar lavage cultures	8.6
7. Faverio, et al. 2016 [7]	Italy	San Gerardo Hospital	2006-2014	65	162	32	PS	Bronchoalveolar /sputum culture	19.8
8. Buscot, et al. 2016 [8]	France	University Hospital	2002-2012	61.0	196	7	RS	Sputum/ bronchoalveolar lavage culture	3.6
9. Dimakou, et al. 2016 [9]	Greece	Hospital of Chest Diseases	2009-2014	60.5	205	2	PS	Sputum culture	0.9
10.* Park, et al. 2016 [10]	**Korea**	**University Hospital >5 years**	**2003-2013**	**59.6**	**155**	**69**	**RS**	**Sputum/BAL culture**	**44.5**
11.* Aksamit, et al. 2017 [11]	**USA**	**Bronchiectasis Research Registry**	**2008-2014**	**66**	**1,314**	**653**	**RS**	**Sputum culture**	**49.7**
12.* Hsieh, et al. 2018 [12]	**China**	**Bronchiectasis Medical Center**	**2005-2014**	**65.3**	**96**	**35**	**RS**	**Sputum culture**	**36.5**
13. Sin, et al. 2019 [13]	Korea	National University Hospital	2005-2016	59.6	6957	1740	RS	Sputum culture	25
14.* Amorim, et al. 2019 [14]	**Portugal**	**Bronchiectasis Referral Center**	**2011-2017**	**54.7**	**186**	**27**	**PS**	**Sputum culture**	**14.5**
15.* Visser, et al. 2019 [15]	**Australia**	**Australian Bronchiectasis Registry**	**2016-2018**	**71**	**169**	**40**	**PS**	**Sputum culture**	**23.6**
16. Pieters, et al. 2019 [16]	Netherland	University Medical Center	2012-2016	60	120	6	RS	Sputum culture	5.0
17.* Dhar, et al. 2019 [17]	**India**	**bronchiectasis registry**	**2015-2017**	**56**	**1299**	**8**	**PS**	**Sputum culture**	**0.6**

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Author; year	Country	Data Source	Time of sampling	Mean age (Y)	Sample size	Patients with NTM	Study design	Method of detection	NTM [%]
18. Huang, et al. 2020 [18]	China	Medical Record Database	2002-2016	65.5	8385	304	RS	Sputum culture	3.6
19. Darwish, et al. 2020 [19]	Egypt;	University Hospital	2017-2018	55.2	40	3	PS	Sputum PCR	7.5
20.* Kwak, et al. 2020 [20]	**Korea**	**Hospital**	**2011-2019**	**62**	**221**	**35**	**PS**	**Sputum culture**	**15.8**
21. Sharif, et al. 2020 [21]	Pakistan	Hospital	2017-2019	NA	196	2	PS	Sputum culture	1.0
22.* Choate, et al. 2021 [22]	**USA**	**Bronchiectasis & NTM Research Registry**	**2008-2018**	**63.7**	**1831**	**885**	**RS**	**Sputum culture**	**48.3**
23.* Metersky, et al. 2021 [23]	**USA**	**Bronchiectasis & NTM Research Registry**	**NA-2020**	**61.1**	**410**	**47**	**PS**	**Sputum culture**	**11.5**
24.Yin, et al. 2021 [24]	China	Hospital	2018-2020	62	202	47	RS	Sputum/BAL culture	23.3

Abbreviations: NA: Not Available; NTM: Nontuberculous Mycobacteria; MBC: mycobacteria culture; AFB: acid fast bacillus smear; BAL: bronchoalveolar lavage; RS: Retrospective Study; PS: prospective Study; UK: United Kingdom; US: United States. PCR: Polymerase Chain Reaction.

Note: Bold type with an asterisk * indicate the studies with source/selection bias. Please refer to the supplementary Table S3.

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Table S3: Ratings of the quality of the evidence and Risk of Bias assessment

Study	Study Design	Domain	Source of bias	Support for Judgement	Review Author's Judgement	Risk of Bias
1. McShane, et al. 2012; US [1]	Retrospective cohort study	Attrition bias	Patient follow-up	Of all 114 patients. Two declined consent, and six patients did not follow up with the evaluation.	Eight patients were not available for the complete analysis.	Low
2. Xu, et al. 2014; Shanghai, China [2]	Retrospective cohort study	Detection bias	Detection bias	Diagnosis criterium was at least one positive culture result of NTM.	Only one positive culture result of NTM may be due to false positive or wrong operation.	Low
3. Guan, et al. 2015; Guangzhou, China [3]	Prospective randomized control cohort study	Selection bias	Patient population	Consecutively recruited from outpatient respiratory clinics.	Mild and severe symptom patients were excluded	Medium
4. Kadowaki, et al. 2015; Japan [4]	Retrospective cohort study	Selection bias	Patient selection	1. Could not perform a full screening for immunodeficiency. 2. Could not eliminate the possibility of cystic fibrosis	1. Patients with immunodeficiency were more vulnerable to NTM infection. 2. CF patients might not be completely excluded.	Low
5.* Máiz, et al. 2016; Spain [5]	**Prospective cohort study**	**Source bias**	**Population selection**	**An observational study of historical cohorts from 4 Spanish teaching hospitals with multidisciplinary and standardized non-CF bronchiectasis outpatient clinics.**	**1. Patients were outpatients, causing severe patients excluded. 2. No unified standard for patient selection.**	**High**
6. Izhakian, et al. 2016; Israel [6]	Retrospective cohort study	Performance bias; Selection bias	Population selection	1.The study precluded patients not under medical care 2.The study is not generalizable to patients treated by other hospital staff members.	1. Patients were in hospital with severe symptoms, causing outpatients excluded. 2. Patients at the same location but from other hospital were not included.	Medium
7. Faverio, et al. 2016; Italy [7]	Prospective cohort study	Selection bias	Population selection	Data were from outpatient clinic and only patients in a stable state were recruited.	Severe patients were excluded for data analysis.	Medium
8. Buscot, et al. 2016; France [8]	Retrospective cohort study	Attrition bias	Follow-up	Functional follow-up was available in 30% of patients with a median duration of 2.7 years.	Only 30% of patients could be followed up, cause attrition bias.	Low
9. Dimakou, et al. 2016; Greece [9]	Prospective cohort study	Detection bias	Detection methods	CF screening, sweat test, saccharin test, and electron microscopy and etc. were employed for patients with different symptoms.	A standard detection method is better for diagnosis and study.	Low

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Study	Design	Bias Domain	Source of bias	Support for Judgement	Review Author's Judgement	Risk of Bias
10.* Park et al. 2016; Korean [10]	**Retrospective cohort study**	**Source bias**	**Patient selection**	**The author only included the Non-CF bronchiectasis patients with followed- up for a minimum of 5 years with CT.**	**The patients were not selected randomly, but have been sick for more than 5 years, causing an overestimate of the prevalence of NTM.**	**High**
11.* Aksamit, *et al.* 2017*[11][11]*; US [11]	**Retrospective cohort study**	**Source bias**	**Data source**	**Data was from Bronchiectasis Research Registry (BRR).**	**The cohort of patients enrolled from tertiary referral institutions with interest in NTM lung disease, the demographic information described is potentially biased, including overrepresentation of patients with NTM.**	**High**
12.* Hsieh, et al. 2018; Taiwan, China [12]	**Retrospective cohort study**	**Source bias; Selection bias**	**Data source and sample size**	**1. All 96 patients were from Linkou Medical Center. Patients with previous pulmonary tuberculosis and those who had received anti-NTM therapy were excluded. 2. Small sample population (<100).**	**Cases from medical center have source bias; Exclusion of some patients causes selection bias.**	**High**
13. Sin, et al. 2019; South Korea [13]	Retrospective controlled cohort study	Detection bias	False positive	They diagnosed by the isolation of NTM from a respiratory specimen at least once.	Isolation of NTM only once may increase the prevalence due to false positive, causing higher infection rates.	Medium
14.* Amorim, et al. 2019; Portugal [14]	**Prospective cohort study**	**Source bias; performance bias**	**Patient selection and sample size**	**1. The data was from a bronchiectasis referral center. 2. The sample size is lower than 200.**	**Cases from a bronchiectasis referral center have source bias; lower sample size increased the prevalence of NTM.**	**High**
15.* Visser, et al. 2019; Australia [15]	**Prospective cohort study**	**Selection bias; Attrition bias**	**Patient selection Incomplete and missing data**	**1. Potentially selects patients with prominent symptoms, more severe disease, and/or a higher prevalence of NTM. 2. predominantly represents non-indigenous patients 3. Patients with missing variable data were removed.**	**1. Severe patients with prominent symptoms may cause a higher prevalence of NTM. 2. Non-indigenous patients did not represent the whole population 3. Data completeness limits the number of participants.**	**High**
16. Pieters et al. 2019; Netherland [16]	Retrospective cohort study	Performance bias	Incomplete data	The majority of patients were seen every 3 months at the outpatient clinic.	Not all the patients were seen every 3 months at the outpatient clinic.	Low

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Study	Design	Bias Domain	Source of bias	Support for Judgement	Review Author's Judgement	Risk of Bias
17.* Dhar, et al. 2019; India [17]	**Prospective cohort study**	**Source Bias**	**Patient selection**	**All the patients were from Indian bronchiectasis registry centers.**	**Patients from registry centers would inevitably cause an overestimation of the prevalence of NTM infections.**	**High**
18. Huang, et al. 2020; Taiwan, China [18]	Retrospective cohort study	Source bias; Detection bias	Patient Selection	1. Inability to completely confirm the bronchiectasis diagnosis. 2. A portion of bronchiectasis diagnoses based on clinical symptoms and chest radiographs, not a reliable diagnostic tool. 3. underestimated the prevalence of immunodeficiency in our cohort.	Some patients might not be bronchiectasis; immunodeficiency may cause high prevalence due to vulnerability.	Medium
19. Darwish, et al. 2020; Egypt [19]	Prospective cohort study	Detection bias Performance bias	NTM were detected by PCR Too small sample size	1. The sputum samples were assessed by PCR, and positive cases did another PCR after 2 months. 2. Sample size 40 < 200	PCR may cause false positive. The only study (of all 13 studies) NTM was not diagnosed by MBC; Small sample size increased prevalence	Medium
20.* Kwak, et al. 2020; South Korea [20]	**Prospective cohort study**	**Source Bias**	**Patient selection**	**Only the patients without NTM infection were included, and then studied their NTM infection afterwards.**	**The patients are not random cases.**	**High**
21. Sharif, et al. 2020; Pakistan [21]	Prospective cohort study	Source bias	Patient selection	This observational cohort study was conducted in the inpatient department only.	Inpatients were usually with severe symptoms, and that may cause overestimation.	Low
22.* Choate, et al. 2021; US [22]	**Retrospective cohort study**	**Source Bias**	**Patient selection**	**All the patients were from the bronchiectasis and NTM Research Registry (BRR).**	**Patients from registry centers would cause an overestimation of the prevalence of NTM infections.**	**High**
23.* Metersky et al. 2021; US [23]	**Prospective cohort study**	**Source Bias**	**Patient selection**	**All the patients were from the bronchiectasis and NTM Research Registry (BRR).**	**Patients from registry centers would cause an overestimation of the prevalence of NTM infections.**	**High**
24. Yin, et al. 2021; China [24]	Retrospective cohort study	Performance bias	Patient information collection	Much patient information was self-reported and recorded.	The self-reported information might be inaccurate and hidden.	low

Note: Bold type with an asterisk * indicate the studies with source/selection bias.

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Table S4: Statistics and calculation of NTM subspecies

Subspecies of NTM	Number of Studies	All NTM Isolated Patient*	Patients with NTM subspecies	% of NTM Subspecies **	McShane	Kadowaki	Máiz	Izhakian	Faverio	Kwak	Darwish	Buscot	Dimakou	Metersky	Yin	% of NTM Isolated patients
MAC	11	255	193	**75.687**	7	28	9	14	24	29	2	7	2	40	31	
M. simiae	3	56	10	**17.857**	1		1	8								
M. gordonae	3	59	7	**11.864**	1		2		4							
M. xenopi	1	9	1	**11.111**	1											
M. fortuitum	4	103	7	**6.796**	1		2	2						2		
M. abscessus	6	208	21	**10.096**			3	3	1	3				1	10	
M. chelonae	6	182	11	**6.044**	1		1	2	2					2	3	
M. lentiflavum	1	18	1	**5.556**			1									
M. kansasii	3	88	5	**4.88**	1				1						3	
M. terrae	1	47	1	**2.128**										1		
M. shimoidei	1	32	1	**3.125**					1							
Undetermined	3	82	6	**7.317**						4	1			1		
Sum (person)	**7**	**1139**			**9**	**26**	**18**	**29**	**32**	**35**	**3**	**7**	**2**	**47**	**47**	
sum (time)					**13**	**28**	**19**	**29**	**33**	**36**	**3**	**7**	**2**	**47**	**47**	
Bronchiectasis Patients					**106**	**147**	**218**	**339**	**162**	**221**	**40**	**196**	**205**	**410**	**202**	**50.712**

Note: Of all 14 included studies, only 9 provided detection methods of NTM subspecies. *The number of all NTM is the sum of all the patients in the studies with the subspecies. **The percentage (%) of NTM infected patients is the sum of all patients with NTM infection divided by the sum of all bronchiectasis patients in all the 9 studies.

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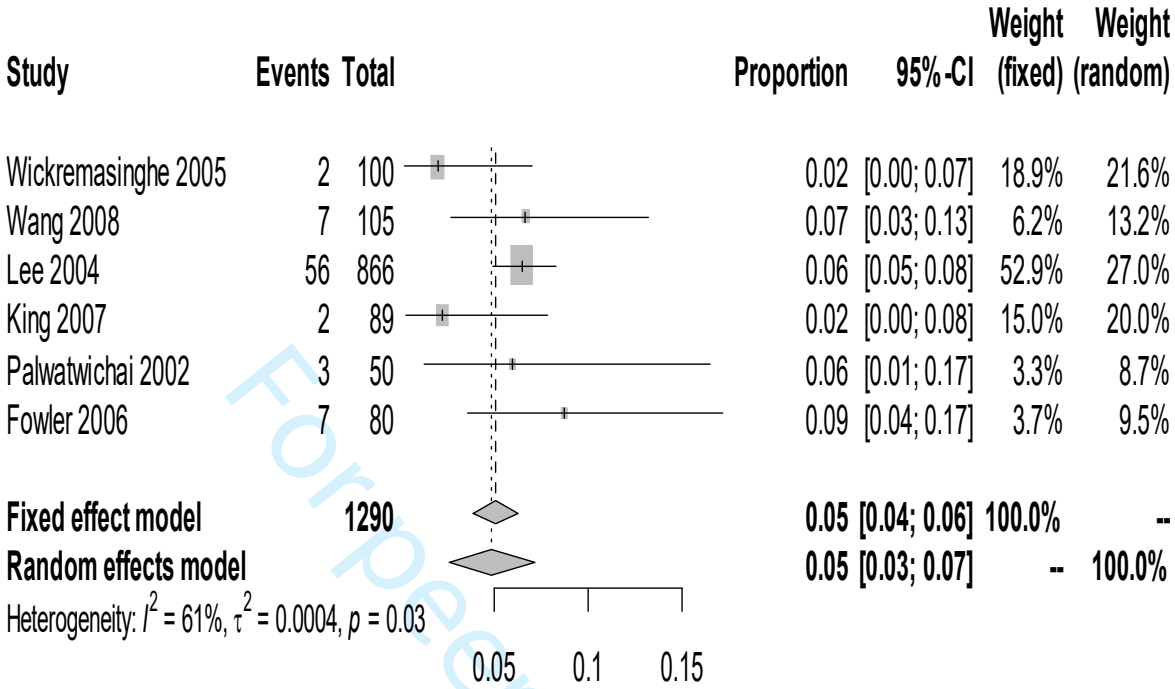


Figure S1: Re-calculating the prevalence of NTM in adults with non-cystic fibrosis bronchiectasis from 1990 to 2006 in Chu's paper "Prevalence of nontuberculous mycobacteria in patients with bronchiectasis: A meta-analysis" published in 2014 [25].

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